

The Crypto–FX Channel into Domestic Equities: Theory and Evidence from Developed and Emerging Markets

Abstract

We study how crypto assets comove with domestic equities across developed and emerging economies. Starting from a tractable portfolio model in which a representative investor allocates between domestic equities and a USD-denominated crypto asset and then repatriates wealth each period, we derive a risk-adjusted parity condition in local-currency terms that links FX-adjusted crypto returns to domestic equity returns through a state-dependent pricing wedge. The model predicts that domestic equities and crypto assets are positively related, but that this comovement can shrink or even reverse when the pricing wedge is large or volatile. This mechanism offers an explanation for why crypto assets such as Bitcoin appeared to diversify or hedge equities in earlier periods, yet increasingly behave as equities as the wedge compresses over time. Using data for eight major developed and emerging economies, we find that comovement is stronger and more persistent in developed markets, consistent with a crypto channel of portfolio integration, but weaker and more heterogeneous in emerging markets, where transmission operates differently. These results are robust to replacing Bitcoin with other major crypto assets and to controlling for standard global risk factors.

1 Introduction

For much of its early history, Bitcoin was heralded as “digital gold,” an uncorrelated asset offering diversification and hedging properties against equities (Brière et al., 2015; Dyhrberg, 2016; Bouri et al., 2017; Kajtazi and Moro, 2019; Stensås et al., 2019, e.g.). Portfolio managers and retail investors alike considered crypto an attractive store of value precisely because its returns appeared disconnected from global stock markets. Indeed, before 2020, rolling correlations between Bitcoin and major equity indices hovered near zero, with box-and-whisker plots from 2017 to 2019 showing medians indistinguishable from non-correlation. This narrative shifted dramatically during the COVID-19 crisis and its aftermath. By 2020, correlations between Bitcoin and the S&P 500 or Nasdaq-100 rose sharply, reaching values near 0.5 and remaining persistently positive thereafter. During episodes of heightened stress (the onset of the pandemic in early 2020, the global tightening cycle of 2022, and the volatility of 2023–2025) Bitcoin and equities moved in lockstep, signaling a structural break from the era when crypto traded as an orthogonal risk factor.

What explains this transition from diversifier to equity-like asset? Several reinforcing factors stand out. First, institutional acceptance has embedded crypto into mainstream portfolios, with asset managers, hedge funds, and even sovereigns (e.g., El Salvador) allocating systematically to digital assets. Second, portfolio integration has deepened comovement, as crypto positions are now rebalanced alongside equities under common risk budgets. Third, volatility dynamics: Bitcoin’s daily volatility remains higher than that of equities, but the volatility wedge has shrunk significantly over the years, magnifying its role as just another high-beta extension of equity exposure. Fourth, supply and holding patterns have matured, with a rising share of coins absorbed by long-term holders and institutions. Finally, increased access through ETFs, futures, and options has further integrated crypto into the global financial ecosystem. These forces collectively align Bitcoin with the global financial cycle, raising the possibility that crypto no longer hedges equities but instead propagates their risks.

The global trend is particularly consequential for both developed and emerging markets (EMs). In advanced economies, crypto has been increasingly embraced as an investable asset class, with institutional allocations expanding through regulated funds and exchange-traded products (e.g., Borri et al., 2025; Magazzino et al., 2025). In EMs, cryptocurrency adoption has accelerated even more strongly, as high and volatile

inflation, recurrent devaluations, shallow financial intermediation, and capital controls broaden the appeal of dollar-linked stores of value and cross-border payments (e.g., Feyen et al., 2022; Murcia et al., 2023; Stupak, 2025). In BRICS economies and other major EMs, flows into crypto assets have grown rapidly, with rising market integration, expanding cross-border stablecoin and Bitcoin transactions, and stronger financial linkages with domestic asset markets (e.g., Auer et al., 2025; Wang et al., 2024; Jangirala et al., 2024; Sam, 2023). Yet the literature offers limited attempts to theoretically explain how crypto markets transmit to domestic asset markets, particularly equities, through the exchange-rate channel, and how this transmission varies across states of the world. For EM investors, crypto assets are not just speculative instruments: because they are denominated in USD, any crypto allocation inherently bundles foreign-exchange exposure with local asset risk. Understanding this linkage is crucial for both prudential policy and portfolio design.

In this paper, we develop and test a model that links three objects that are central to investors: domestic equity returns, USD crypto returns, and the exchange rate. A representative investor allocates wealth between a local equity and a crypto asset, repatriating proceeds into domestic currency each period. This setup yields a natural risk-adjusted parity between local equity returns and a USD-crypto leg converted via the exchange rate. We ask three questions. *First*, how closely does this parity hold in returns, and does the strength of the linkage differ systematically between the developed and emerging economies in our sample? *Second*, how does the wedge behave in bad times when global risk rises, liquidity tightens, or capital controls bind? *Third*, through which channel—crypto innovations or exchange-rate shocks—do disturbances propagate into local equities, and how heterogeneous is this propagation across countries?

Our main contributions are threefold. (1) We formalize a *crypto-FX-equity linkage* and derive propositions that are straightforward to test. (2) We model and measure a *state-dependent wedge* in crypto-equity comovement whose volatility has shrunk over time, offering a structural explanation for the observed shift from low to high comovement. (3) We decompose transmission into *crypto vs. FX channels*, documenting economically meaningful cross-country heterogeneity in how shocks are absorbed by exchange rates versus direct crypto loadings. These findings are robust to adding standard global and macro-financial controls and highlight the channels through which crypto exposure permeates EM financial systems.

Our paper contributes to several strands of literature. Early empirical studies examine the distributional properties, volatility dynamics, and hedging attributes of Bitcoin and other crypto assets (e.g., Urquhart, 2016; Dyhrberg, 2016; Baur et al., 2018; Corbet et al., 2018). This work documents that Bitcoin exhibits high kurtosis, volatility clustering, and fragile hedging performance. For instance, Urquhart (2016) finds significant autocorrelation and departures from weak-form efficiency in early Bitcoin trading, while Dyhrberg (2016) shows that Bitcoin behaves partly like gold and partly like a currency in GARCH specifications. Baur et al. (2018) and Corbet et al. (2018) conclude that Bitcoin is neither a reliable safe haven nor a consistent diversifier, with correlations that spike in periods of market stress. We extend this literature by providing a theoretical mechanism that explains why correlations between Bitcoin and equities were historically low but rose sharply after 2020. Rather than treating the breakdown of the “digital gold” narrative as a statistical artifact, we derive a crypto-FX-equity parity condition that embeds USD-crypto payoffs into domestic returns and generates a state-dependent wedge whose variance we show is central for the crypto-equity slope; empirically, we find that in the post-2020 period the wedge is tighter and the slope is closer to one, consistent with a decline in effective wedge variance as crypto became integrated into global financial markets.

A second strand of research analyzes the pricing of crypto assets, their factor exposures, and their comovement with global risk conditions. Liu and Tsyvinski (2018) show that standard equity factors such as size, value, and momentum do not price Bitcoin, whereas crypto-specific characteristics related to investor attention and supply dynamics do. Yermack (2015) highlights the speculative and commodity-like behavior of cryptocurrencies, noting that their extreme exchange-rate volatility undermines their usefulness as currencies. Borri (2019) documents substantial downside tail risk in Bitcoin and Ethereum, with tail exposure intensifying during global liquidity stress. These studies reveal that crypto assets are increasingly sensitive to global risk but do not provide a theoretical rationale for the rising correlation between crypto returns and domestic equities. Our paper contributes by showing that any USD-denominated crypto payoff mechanically maps into local-currency equity returns through the exchange rate. This delivers a structural explanation for the growing crypto-equity beta and allows us to decompose comovement into a *crypto-innovation channel* and an *FX-transmission channel*, filling a gap in the existing literature.

A growing adoption literature emphasizes the rapid diffusion of crypto assets in emerging markets (EMs), driven by inflation hedging motives, currency instability, capital controls, shallow financial intermediation,

and limited access to formal banking (e.g., Feyen et al., 2022; Copestake et al., 2023; Murcia et al., 2023). Stablecoins such as USDT and USDC have become central to cross-border payments and savings vehicles in dollar-constrained EM environments (e.g., Arner et al., 2020). Market-structure studies highlight the importance of on/off-ramps, crypto-banking linkages, and regulatory frictions for shaping EM participation (e.g., Auer et al., 2025; Jangirala et al., 2024). Yet this literature rarely links USD-crypto exposure to *domestic asset prices*, and provides no theoretical account of how EM equities respond to crypto shocks. Our contribution is to embed crypto adoption into a portfolio-choice framework in which investors repatriate USD-crypto returns into local currency each period. This mechanism naturally transmits crypto innovations and FX movements into domestic equity pricing, helping explain why emerging markets exhibit more FX-dominated and state-dependent crypto-equity comovement than advanced economies, where the linkage is tighter and more equity-like.

Our framework is closely connected to classic research on exchange-rate exposure and pass-through. Adler and Dumas (1984) establish the foundational model in which equity prices respond to currency mismatches in firms’ cash flows, while Jorion (1990) provides early empirical evidence on the heterogeneity of exchange-rate betas for U.S. multinationals. These studies show that foreign-currency payoffs translate mechanically into domestic-asset returns. We extend this canonical framework by treating crypto assets as USD-denominated instruments whose local-currency value depends on the exchange rate. Rather than focusing on cash flows or trade exposure, our model highlights a *portfolio-driven* pass-through mechanism: the domestic pricing implications of holding USD-crypto assets. This reframes crypto not as an isolated asset class but as an alternative channel of foreign-currency exposure embedded in domestic portfolios.

Our analysis also relates to the global financial cycle literature, which emphasizes how U.S. monetary policy, global risk appetite, and dollar funding conditions drive co-movement across international asset markets. Hau and Rey (2006) show that global equity flows affect currency movements through intermediary rebalancing, while Bruno and Shin (2015) highlight how fluctuations in dollar funding and leverage transmit global liquidity conditions to EMs. These ideas underpin the view that the dollar plays a dominant role in shaping cross-border asset dynamics. We extend this perspective by incorporating crypto assets into the global-financial-cycle framework: because Bitcoin and other major crypto assets are effectively USD-denominated, shocks to crypto valuations act as additional conduits for dollar risk to propagate into domestic equity markets. Our empirical decomposition quantifies how much of local equity variation is attributable to FX pass-through versus crypto-specific shocks, demonstrating the emergence of a “crypto-enhanced” dollar cycle.

Finally, research on capital controls, liquidity constraints, and segmented financial markets shows that the degree of USD transmission into domestic assets depends on institutional frictions (e.g., Adler and Dumas, 1983; Jorion, 1990; Hau and Rey, 2006; Bruno and Shin, 2015). When capital controls bind, FX markets are shallow, or dollar liquidity is scarce, exchange-rate pass-through into domestic asset prices rises (e.g., Rey, 2015; Lovchikova and Matschke, 2024). Although this literature has not considered crypto assets explicitly, crypto trading and stablecoin usage frequently circumvent conventional capital-account restrictions and interact with FX markets in countries facing currency pressure (e.g., Murcia et al., 2023). We contribute by showing that the crypto-FX-equity linkage is amplified in precisely these environments. In countries with tighter capital controls or thinner FX markets, the FX component of USD-crypto returns dominates the transmission mechanism, making domestic equities more sensitive to crypto cycles. This reveals a policy-relevant channel through which digital assets function as a shadow conduit of dollar exposure in partially closed economies.

The rest of the paper is organized as follows. In Section 2, we discuss the stylized facts that captures the main idea documented in this paper. Section 3 presents the model and its implications and develops testable propositions. Section 4 outlines the empirical strategy and data. We also report the main results and cross-country heterogeneity. Section 5 concludes.

2 Stylized Fact for Intuition to Build Motivation

To build intuition for the main idea documented in this paper, we first explore how the comovement between stock market returns and Bitcoin returns has evolved since the advent of Bitcoin trading, and how simple wedges between these two markets have developed over time. Specifically, we track (i) the rolling correlation

of Bitcoin and equity returns, (ii) a *returns wedge* defined as the difference between Bitcoin and equity returns, and (iii) a *volatility wedge* defined as the difference between their realized volatilities. These are purely descriptive objects, distinct from the structural risk-adjusted wedge in our model, but they provide useful intuition about whether the two asset classes remain distinct or increasingly behave as a single high-beta equity factor.

As shown in Figure 1, Bitcoin initially exhibited little to no comovement with equities. The one-year rolling correlation of returns hovered near zero or negative values, supporting the early perception that Bitcoin could act as a hedge against equities. This period coincided with elevated volatility in both the returns wedge and the volatility wedge: Bitcoin’s idiosyncratic risk profile dominated, creating a clear separation from the stock market and reinforcing its appeal as a diversification instrument.

Between 2016 and 2019, the volatility of these wedges began to decline markedly. As wedge volatility compressed, the correlation of Bitcoin with equities rose gradually, with several episodes of positive and statistically significant comovement. This suggests that the distinctiveness of Bitcoin as an independent asset class started to erode as market forces began aligning it more closely with equity performance. Post-2019, the patterns become striking. The volatility wedges shrank to historically low levels, while the correlation between Bitcoin and equities surged—at times exceeding 0.55 on a one-year rolling basis. Monthly rolling correlations (not shown) occasionally surpass 0.90, indicating that Bitcoin and equity markets are now deeply intertwined. This convergence implies that Bitcoin has lost much of its hedging or diversification property and now often amplifies equity market dynamics rather than offsetting them. The COVID-19 shock in 2020 and the 2022 monetary tightening cycle provide clear stress tests. In both cases, correlations spiked unusually fast, highlighting Bitcoin’s increasing tendency to transmit, rather than diversify, systemic shocks. These crisis dynamics reinforce the broader finding: Bitcoin now behaves more like a high-beta equity factor than as a diversifier.

The evidence points to a shift: Bitcoin’s behavior is no longer governed by idiosyncratic drivers alone. As wedge volatility declined, the “idiosyncratic gap” between Bitcoin and equities diminished, and both assets increasingly load on common factors. This convergence is unlikely to be explained by differences in pricing kernels, since both assets are priced under broadly similar stochastic discount factors. Instead, the data suggest that the compression of wedge volatility has eroded Bitcoin’s uniqueness as an asset class, making its return dynamics nearly indistinguishable from equities. From a market perspective, this stylized fact is crucial: the narrative of Bitcoin as “digital gold” or a hedge is increasingly untenable. Instead, Bitcoin’s evolution suggests it now acts as a high-beta component of the equity market, at least relative to the past years. For consistency, we use the highest possible interday frequency—daily returns—and compute one-year rolling windows to smooth short-term noise while retaining the visibility of the documented shifts. Although not reported here, alternative specifications using monthly windows reinforce the same conclusion. In the next section, we present a model that links Bitcoin and equity comovement.

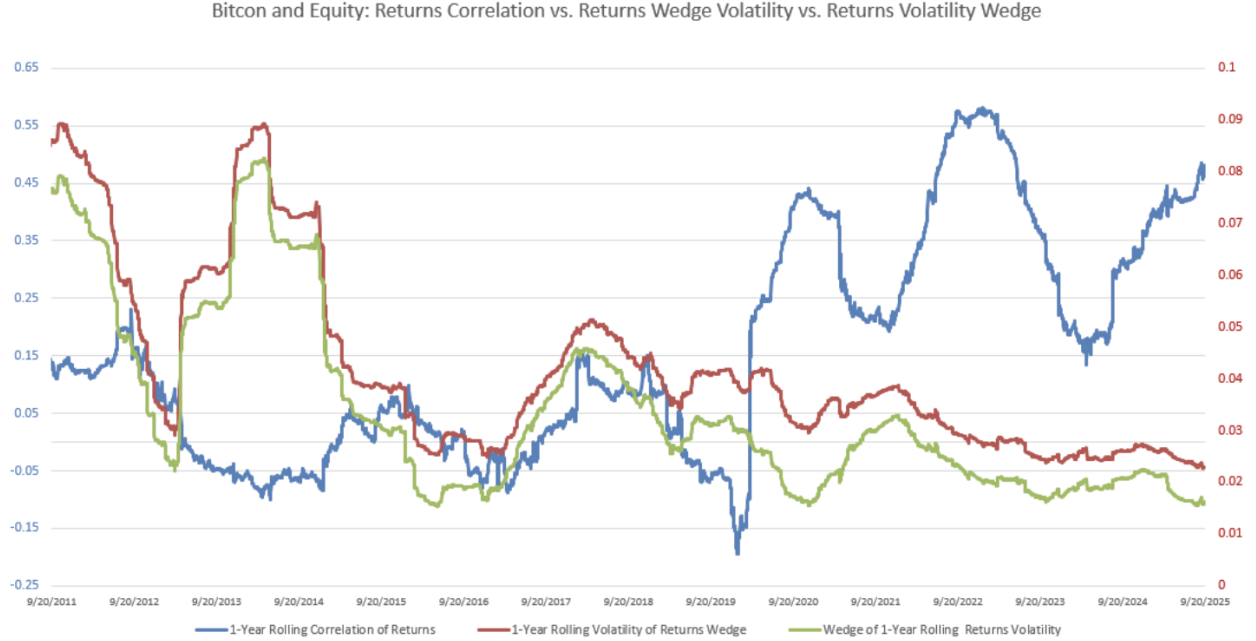


Figure 1: **Bitcoin and Equity:** 1-year rolling correlation of returns (left axis; blue) versus the volatility of the returns wedge (red) and the wedge of 1-year rolling returns volatility (green; right axis). The figure illustrates three phases: an early period of weak or negative comovement with high wedge volatility (2011–2016), a transition with compressing wedges and rising comovement (2016–2019), and a post-2019 convergence where correlations become persistently positive and high. Crisis episodes such as COVID-19 (2020) and the 2022 tightening cycle further reinforced Bitcoin’s equity-like behavior.

3 A Model of Equity and Crypto Relationship

Motivated by the reduced-form correlations, the volatility of returns, and the wedge evidence presented in the previous section, we now build a stylized model in which agents hold positions in a domestic equity and in Bitcoin (representative of a USD-denominated crypto asset). Because the investor’s wealth is expressed in the domestic currency (LCU), whenever the unit of account differs from the U.S. dollar we must allow the exchange rate to enter the model explicitly. When flows are dollar-denominated or when the domestic currency *is* the dollar, we set the exchange rate to one without loss of generality. Section 3.1 presents the model setup. Section 3.2 outlines the budget constraint faced by the investor. Section 3.3 states the associated Bellman equation and solves the first-order conditions. Section 3.4 discusses the propositions implied by the model. Section 3.5 outlines potential extensions.

3.1 Model Setup

To concretize the idea in a tractable environment, consider a representative domestic investor who chooses consumption and portfolio holdings in discrete periods $t = 0, 1, 2, \dots$. The investor can hold two risky assets: a domestic stock market claim and a global crypto asset (e.g., Bitcoin) that is priced in USD. At the end of each period, all wealth is marked to local currency (LCU). The investor receives exogenous labor income I_t in LCU. Let C_t denote consumption and P_t^c the domestic price level (CPI, in LCU). The gross inflation rate is :

$$1 + \pi_{t+1} = \frac{P_{t+1}^c}{P_t^c}.$$

For the domestic stock market, let P_t^s be the stock price (LCU) and D_t the dividend (LCU) paid at t so that D_{t+1} is the dividend (LCU) paid at $t+1$. The gross *nominal* and *real* returns are

$$1 + R_{t+1}^s = \frac{P_{t+1}^s + D_{t+1}}{P_t^s}, \quad 1 + R_{t+1}^{s,\text{real}} = \frac{1 + R_{t+1}^s}{1 + \pi_{t+1}}.$$

For the cryptocurrency asset priced in USD, let Θ_{t+1}^* be the USD return on the crypto asset (e.g., BTC) and define its log return $\theta_{t+1} \simeq \log(1 + \Theta_{t+1}^*)$. Let S_t be the domestic exchange rate expressed as LCU per USD. The FX log return is

$$\Delta s_{t+1} = \log S_{t+1} - \log S_t,$$

where $\Delta s_{t+1} > 0$ means the dollar has appreciated against the LCU. The investor chooses (i) the number of domestic shares n_t to carry from t to $t+1$ and (ii) the LCU amount A_t^k to allocate to the crypto leg at t . Investing A_t^k converts to a USD amount A_t^k/S_t at t , earns USD gross return $(1 + \Theta_{t+1}^*)$ over $[t, t+1]$, and is repatriated at S_{t+1} into LCU at $t+1$.

3.2 Budget Constraint

The investor enters period t with positions (n_{t-1}, A_{t-1}^k) . After receiving all inflows at t , she chooses (C_t, n_t, A_t^k) and carries only the two risky legs (equity and crypto) into $t+1$. Thus, in LCU, **Inflows at t :**

$$\underbrace{(P_t^s + D_t) n_{t-1}}_{\text{stock sale + dividend}} + \underbrace{\frac{S_t}{S_{t-1}} (1 + \Theta_t^*) A_{t-1}^k}_{\text{crypto proceeds in LCU after FX}} + \underbrace{I_t}_{\text{labor income}}.$$

Outflows at t :

$$\underbrace{P_t^c C_t}_{\text{consumption}} + \underbrace{P_t^s n_t}_{\text{new stock position}} + \underbrace{A_t^k}_{\text{new crypto allocation}}.$$

Hence the flow budget constraint is

$$P_t^c C_t + P_t^s n_t + A_t^k = (P_t^s + D_t) n_{t-1} + \frac{S_t}{S_{t-1}} (1 + \Theta_t^*) A_{t-1}^k + I_t. \quad (1)$$

Define the LCU gross nominal and real returns on the crypto leg:

$$1 + R_{t+1}^k := \frac{S_{t+1}}{S_t} (1 + \Theta_{t+1}^*), \quad 1 + R_{t+1}^{k,\text{real}} = \frac{1 + R_{t+1}^k}{1 + \pi_{t+1}}. \quad (2)$$

In logs (first-order approximation), $r_{t+1}^k \approx \theta_{t+1} + \Delta s_{t+1}$.

3.3 Investor's Problem and Risk-Adjusted Parity

Following standard arguments, we assume preferences are time-separable with discount factor $0 < \beta < 1$ and strictly increasing, concave $u(\cdot)$. The investor chooses $\{C_t, n_t, A_t^k\}_{t \geq 0}$ to

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t) \quad \text{s.t.} \quad (1), \quad n_t \geq 0, \quad A_t^k \geq 0. \quad (3)$$

Define cash-on-hand (LCU) at time t :

$$M_t := (P_t^s + D_t) n_{t-1} + \frac{S_t}{S_{t-1}} (1 + \Theta_t^*) A_{t-1}^k + I_t.$$

The investor chooses (C_t, n_t, A_t^k) to solve the Bellman problem

$$V(M_t) = \max_{C_t, n_t, A_t^k} \left\{ u(C_t) + \beta \mathbb{E}_t [V(M_{t+1})] \right\}$$

subject to

$$P_t^c C_t + P_t^s n_t + A_t^k = M_t, \quad M_{t+1} = (P_{t+1}^s + D_{t+1}) n_t + \frac{S_{t+1}}{S_t} (1 + \Theta_{t+1}^*) A_t^k + I_{t+1}.$$

Let $V_M(\cdot)$ denote the derivative of V with respect to cash-on-hand. The standard first-order conditions yield to Euler equations in real LCU. Let the real pricing kernel (stochastic discount factor) be

$$m_{t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

Lemma 1 in Appendix A derives its exact form. Then

$$\mathbb{E}_t[m_{t+1} (1 + R_{t+1}^s)] = 1, \quad \mathbb{E}_t[m_{t+1} (1 + R_{t+1}^k)] = 1,$$

and, equivalently, the real gross returns on the domestic equity leg and the FX-converted crypto leg satisfy in real terms,

$$\mathbb{E}_t[m_{t+1} (1 + R_{t+1}^{s,\text{real}})] = 1, \quad \mathbb{E}_t[m_{t+1} (1 + R_{t+1}^{k,\text{real}})] = 1.$$

Subtracting the two conditions, we obtain the *risk-adjusted parity*:

$$\mathbb{E}_t \left[m_{t+1} \left((1 + R_{t+1}^s) - (1 + R_{t+1}^k) \right) \right] = 0 \iff \boxed{\mathbb{E}_t[m_{t+1} (R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}})] = 0}.$$

Defining the parity wedge

$$w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}},$$

$$\mathbb{E}_t[m_{t+1} w_{t+1}] = 0.$$

Using the identity

$$\mathbb{E}_t[m_{t+1} w_{t+1}] = \mathbb{E}_t[m_{t+1}] \mathbb{E}_t[w_{t+1}] + \text{Cov}_t(m_{t+1}, w_{t+1}),$$

the parity condition implied that

$$E_t[w_{t+1}] = - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]}. \quad (4)$$

Decompose the wedge w_{t+1} around its conditional mean:

$$w_{t+1} = E_t[w_{t+1}] + \varepsilon_{t+1}, \quad E_t[\varepsilon_{t+1}] = 0. \quad (3)$$

$$\varepsilon_{t+1} := w_{t+1} - E_t[w_{t+1}]. \quad (5')$$

Since $R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} + w_{t+1}$, it follows that

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \quad E_t[\varepsilon_{t+1}] = 0. \quad (4)$$

Thus, in our setup both legs are priced by the same real SDF. Any deviation between the domestic equity return and the FX-converted crypto return can be summarized by a real parity wedge $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$ that satisfies the risk-adjusted parity $\mathbb{E}_t[m_{t+1} w_{t+1}] = 0$. Proposition 1 below shows that the *conditional mean* of this wedge is entirely pinned down by its covariance with the SDF, and Proposition 2 characterizes how the volatility of the wedge constrains the short-run regression slope between the two legs.

Thus, in our setup both legs are priced by the same real SDF, and any systematic deviation between the domestic equity return and the FX-converted crypto return must be compensated by a *priced wedge* w_{t+1} that is orthogonal to the SDF in expectation. In Section 3 we summarize the implications of this wedge structure for the conditional mean and for the short-run regression slope; the formal statements and proofs are collected in Propositions 1–2 and Appendix A.

Corollary Under the model condition, projecting $R_{t+1}^{s,\text{real}}$ on $R_{t+1}^{k,\text{real}}$ yields

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1}, \quad E_t[u_{t+1}] = 0,$$

where α captures the β measures the strength of comovement, and u_{t+1} is the mean-zero wedge innovation. The model predicts $\beta > 0$ and $\beta \rightarrow 1$ as the idiosyncratic wedge variance vanishes. The structural parity condition implies

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \quad E_t[\varepsilon_{t+1}] = 0,$$

so the conditional mean of the wedge is determined by the interaction of the pricing kernel m_{t+1} with the wedge w_{t+1} . In the empirical regression

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1},$$

this adjustment cannot be observed directly. Its average effect is absorbed into the intercept α , while any time-varying deviations not captured by α appear in the residual u_{t+1} . Thus α and u_{t+1} together proxy for the SDF–wedge term in empirical implementation. We now generate the main testable proposition linking the domestic equity and crypto assets. We now state two testable propositions that link the domestic equity leg and the FX-converted crypto leg implied by the first order condition.

We now state two testable propositions implied by the model: one on the priced wedge decomposition and one on the short-run slope

3.4 Propositions

Proposition 1 (wedge decomposition).

Let $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$ denote the (real) parity wedge between the domestic equity leg and the FX-converted crypto leg. Under the Euler equations in §3.3, the wedge satisfies

$$\mathbb{E}_t[w_{t+1}] = - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]}.$$

Moreover, writing $w_{t+1} = \mathbb{E}_t[w_{t+1}] + \varepsilon_{t+1}$ with $\mathbb{E}_t[\varepsilon_{t+1}] = 0$, we have

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} + \varepsilon_{t+1},$$

so that the nonzero conditional mean of the wedge is entirely driven by the interaction between the pricing kernel m_{t+1} and the return differential w_{t+1} . In particular, if $\text{Cov}_t(m_{t+1}, w_{t+1}) = 0$, then $\mathbb{E}_t[R_{t+1}^{s,\text{real}}] = \mathbb{E}_t[R_{t+1}^{k,\text{real}}]$.

Note: In empirical implementation we do not observe the SDF–wedge term $(-\text{Cov}_t(m_{t+1}, w_{t+1})/\mathbb{E}_t[m_{t+1}])$ directly. Instead, we project $R_{t+1}^{s,\text{real}}$ on $R_{t+1}^{k,\text{real}}$,

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1}, \quad \mathbb{E}_t[u_{t+1}] = 0.$$

Comparing this regression with Proposition 1 makes clear that the average effect of the priced wedge is absorbed into the intercept α , while any time-varying component of the SDF–wedge interaction that is not captured by α appears in the residual u_{t+1} . Thus α and u_{t+1} together proxy for the priced wedge in our empirical specification, and Proposition 2 below characterizes the resulting short-run slope β .

Proposition 2 (short-run slope $> 0, \approx 1$).

In the short run, after controlling for common shocks,

$$r_{t+1}^{s,\text{real}} = \alpha + \beta r_{t+1}^{k,\text{real}} + \lambda_{t+1} + u_{t+1}.$$

with time fixed effects as the controls for common shocks and the time fixed effects λ_{t+1} span the space of F_{t+1} generated by the common shocks, i.e., where F_{t+1} stacks the common risk shocks.

Let ε_{t+1}^s and ε_{t+1}^k be the idiosyncratic components of $r_{t+1}^{s,\text{real}}$ and $r_{t+1}^{k,\text{real}}$ after removing common shocks, and define the idiosyncratic parity wedge $e_{t+1} := \varepsilon_{t+1}^s - \varepsilon_{t+1}^k$. Then

$$\beta = \frac{\text{Cov}(\varepsilon_{t+1}^s, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)} = 1 + \frac{\text{Cov}(e_{t+1}, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)},$$

and by Cauchy–Schwarz,

$$1 - \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^k)}} \leq \beta \leq 1 + \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^k)}}.$$

In particular, if the wedge variance is small, the slope is strictly positive and close to one.

For convenience, define the **square-root variance ratio**

$$\sqrt{\rho} := \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^k)}},$$

so that the bounds can be written compactly as $1 - \sqrt{\rho} \leq \beta \leq 1 + \sqrt{\rho}$. In particular, if $\text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^k) < 1$, then $\beta > 0$; moreover, as $\text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^k) \rightarrow 0$, we have $\beta \rightarrow 1$ as required.¹

The intuition here is that both legs are priced by the same SDF; when the idiosyncratic wedge e_{t+1} is small relative to the volatility of the FX-converted crypto leg, the regression slope is forced to lie strictly above zero and concentrate near one. The inequality quantifies this: the tighter the residual wedge, the tighter β is around one. In other words, crypto and domestic equities are positively linked and comove in the short run, rather than offsetting each other. The smaller is the variance of their idiosyncratic wedge, the stronger is the relationship between crypto and equities. This also suggests that periods where equities and crypto returns have offset each other or exhibited very weak links are periods characterized by large wedge variance, driven by factors such as capital controls, liquidity freezes, or policy interventions. Previously, Bitcoin and equities appeared unrelated because the wedge variance was large: thin liquidity, fragmented markets, and regulatory frictions meant crypto returns deviated significantly from equity-linked parity, widening the bounds around the slope β and making estimated correlations indistinguishable from zero. More recently, large institutional, individual, or even sovereign adoption, portfolio integration, and ETF/futures access compressed wedge variance, with the relationship concentrating at higher values and explaining why correlations jumped and stayed persistently positive. Thus, both crypto and equities load on the same stochastic discount factor, and the smaller the wedge variance, the stronger both assets comove and the closer the slope is to one. Our model shows that crypto has therefore shifted from offsetting equity risk to transmitting it—a move from the “digital gold” narrative to equity-like behavior.

3.5 Extensions

The baseline model is intentionally stylized, designed to isolate the core equity–crypto relationship through the FX channel. Nevertheless, several natural extensions could be considered. First, one could introduce heterogeneity in investor types, distinguishing between financially integrated households and liquidity-constrained agents. This would allow wedge variance to depend on market participation and access, potentially explaining cross-country heterogeneity. Second, relaxing the assumption of frictionless markets to incorporate capital controls, liquidity freezes, or transaction costs would make explicit the mechanisms by which wedge variance spikes during stress. Third, the model could be extended to multiple crypto assets (e.g., Bitcoin and Ether), or to stablecoins alongside volatile tokens, to capture a richer menu of USD-linked instruments. Importantly, none of these extensions would overturn the main result of Proposition 2: as long as both legs load on the same SDF, the short-run slope remains strictly positive and close to unity whenever the idiosyncratic wedge variance is small. We now move to the empirical analysis section.

¹Throughout, R denotes a gross return and $r := \log(1 + R)$ its log return. The regression specification is stated in log returns for tractability, but it is directly implied by the structural relation in gross returns.

4 Empirical Evidence and Analysis

This section presents the empirical results. During the Post-2020 period, in major developed markets, FX-adjusted crypto returns and local equity indices generally move positively together in a stable, equity-like manner, and this comovement remains strong after controlling for standard global risk factors. While in emerging markets, the relationship is present but weaker and more uneven, with much of the connection appearing in country-specific rather than a single tight loading for each country.

4.1 Empirical Strategy and Data

In this section we take the risk-adjusted parity condition in Section 3 to the data. The model implies that, after converting the USD crypto leg into local currency and working in real terms, domestic equity returns should load positively on the FX-adjusted crypto return, with the short-run slope β strictly positive and approaching one as the idiosyncratic wedge variance shrinks (Proposition 2). Our empirical strategy is organized around three objects: (i) a baseline panel regression of local equity returns on FX-adjusted crypto returns; (ii) a decomposition into crypto and FX channels; and (iii) an analysis of how the wedge behaves across states of the world, in particular during periods of global stress and tight financial conditions.

4.2 Data and Construction of Returns

We assemble a panel of eight major developed and emerging economies. For each country i we obtain: a broad, value-weighted equity index in local currency (LCU), the bilateral nominal exchange rate against the U.S. dollar. For the crypto leg, we use Bitcoin as the baseline USD-denominated asset and later verify robustness using other major crypto assets (e.g., Ether, a broad crypto index). All series are sampled at the highest common frequency available, daily in our main specification, and we also consider other crypto returns for robustness.

Let $P_{i,t}^s$ denote the equity index level (LCU) for country i at time t , and let P_t^k denote the USD price of the crypto asset. Let $S_{i,t}$ be the bilateral spot exchange rate, expressed as LCU per USD so that an increase in $S_{i,t}$ corresponds to a local-currency depreciation (dollar appreciation). We define log returns over horizon h as:

$$r_{i,t+1}^s := \log P_{i,t+1}^s - \log P_{i,t}^s, \quad r_{t+1}^{k,USD} := \log P_{t+1}^k - \log P_t^k, \\ \Delta s_{i,t+1} := \log S_{i,t+1} - \log S_{i,t}.$$

The FX-adjusted crypto return in local currency is then

$$r_{i,t+1}^{k,LCU} := r_{t+1}^{k,USD} + \Delta s_{i,t+1},$$

which is the log return that a domestic investor earns on the crypto leg after converting USD proceeds back into LCU. In the strict model, both $r_{i,t+1}^s$ and $r_{i,t+1}^{k,LCU}$ should be measured in real terms. Since our horizons are short (daily and weekly), we treat nominal log returns as close empirical counterparts to real returns; in robustness checks at monthly horizons, we subtract CPI inflation to construct real returns.

We split the sample into advanced and emerging economies using standard classifications and report all results separately for these two groups. This allows us to assess whether the crypto–FX channel into local equities is stronger and more stable in developed markets and more heterogeneous or FX-dominated in emerging markets, as suggested by the model and the stylized facts in Section 2.

4.3 Baseline Specification: Crypto–Equity Parity in Local Currency

The core implication of Proposition 2 is that, after controlling for common shocks, the short-run slope of domestic equity returns on FX-adjusted crypto returns should be strictly positive and approach one as the idiosyncratic wedge variance becomes small. To test this prediction, we estimate the following panel regression:

$$r_{i,t+1}^s = \alpha_i + \beta r_{i,t+1}^{k,LCU} + \Gamma' Z_{i,t+1} + \varepsilon_{i,t+1}, \quad (5)$$

where $r_{i,t+1}^s$ is the local equity log return (LCU) for country i , $r_{i,t+1}^{k,LCU}$ is the FX-adjusted crypto return, α_i is a country fixed effect, and λ_t denotes time fixed effects that absorb common shocks and span the space of

aggregate risk factors F_{t+h} in the model. The vector $Z_{i,t+h}$ collects control variables discussed below, and $\varepsilon_{i,t+h}$ is the residual, which we interpret as the empirical counterpart of the wedge innovation u_{t+1} in the model.

The parameter of interest is β . Under the model, $\beta > 0$ and $\beta \rightarrow 1$ as the variance of the idiosyncratic parity wedge e_{t+1} shrinks relative to the variance of the idiosyncratic crypto component (Proposition 2). Empirically, we interpret larger values of β as evidence of a tighter crypto–equity linkage in local currency. Comparing estimates of (5) across advanced and emerging markets then provides a direct test of whether this linkage is stronger and more stable in developed markets, and whether EMs exhibit weaker or more state-dependent transmission.

The vector $Z_{i,t+h}$ contains controls. We follow the global risk and intermediary-constraint literature and include the change of the VIX, to capture the global risk sentiment. We include macro-financial controls capturing monetary and yield-curve conditions: the change in the U.S. term spread (10-year minus 2-year Treasury yield), and the change in the term-spread differential between country i and the U.S. These controls help absorb slow-moving differences in monetary policy stances that might jointly affect equity and FX returns. We also include commodity returns as controls.

4.4 United States

In this subsection, we study whether USD-denominated crypto assets, once converted into local currency, move systematically with domestic equity markets.

Tables 1 and 2 present daily regressions of U.S. equity returns on USD-denominated Bitcoin returns, estimated separately for the pre-2020 and post-2020 samples. The specifications progressively incorporate controls for commodities, the dollar, implied volatility, and the yield curve, thereby isolating the incremental role of Bitcoin.

Table 1: Pre-2020 United States: Bitcoin–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{BTC}_{USD})$	0.005** (0.002)	0.003 (0.002)	0.003 (0.002)	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.000)	-0.002*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
ΔVIX				-0.005*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.025*** (0.002)	0.025*** (0.002)
$\Delta 1\text{M Yield}$						-0.001 (0.004)
Observations	3451	3451	3451	3451	3451	3451
Adj. R^2	0.001	0.122	0.146	0.721	0.737	0.737

Note: Dependent variable is the U.S. equity index return. Robust SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The pre-2020 estimates in Table 1 reveal that Bitcoin had almost no systematic connection to U.S. equities. In the bivariate specification, the slope on $\Delta \log(\text{BTC}_{USD})$ is small and only marginally significant (0.005 in column (1)). As controls for common macro-financial shocks are introduced, the coefficient declines and eventually turns slightly negative (around -0.002 in columns (4) to (6)), while the adjusted R^2 rises sharply from 0.001 to approximately 0.74. This pattern indicates that prior to 2020, broad macro financial forces (movements in commodities, the dollar, volatility, and interest rates) accounted for most variation in U.S. equities, leaving Bitcoin with negligible incremental explanatory power once these drivers were absorbed. Conceptually, this reflects a period characterized by a large parity wedge: crypto markets remained thin, fragmented, and only loosely connected to the U.S. pricing kernel, preventing Bitcoin from loading meaningfully on the global dollar risk environment.

Table 2: Post-2020 United States: Bitcoin–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{BTC}_{USD})$	0.135*** (0.007)	0.125*** (0.007)	0.114*** (0.007)	0.036*** (0.005)	0.036*** (0.005)	0.036*** (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.000*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.007*** (0.001)	-0.004*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.008*** (0.003)
$\Delta 1\text{M Yield}$						0.003 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.145	0.177	0.212	0.646	0.647	0.647

Note: Dependent variable is the U.S. equity index return. Robust SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

By contrast, the regressions for the post-2020 period in Table 2 reveal a pronounced strengthening of the crypto–equity linkage. The slope on $\Delta \log(\text{BTC}_{USD})$ is large and highly significant in the bivariate specification (0.135 in column (1)), and although attenuated by controls, it remains economically meaningful and statistically robust at ≈ 0.036 in columns (4) to (6). The adjusted R^2 rises from roughly 0.15 to about 0.65 across specifications. This shift aligns with the model’s prediction that as the parity wedge compresses, both assets load on the same pricing kernel. The square root variance ratio further supports this interpretation: while both periods lie close to unity, the post-2020 value is slightly smaller ($\sqrt{\rho} \approx 0.990$ compared with 1.005 before 2020), signaling tighter parity and a reduced residual gap.

Taken together, these results point to a structural reclassification of Bitcoin within the global dollar asset ecosystem. After 2020, Bitcoin becomes increasingly integrated into the risk environment that prices U.S. equities. As crypto markets deepen, gain institutional participation, and become more connected to dollar funding markets, Bitcoin begins to move in tandem with the global financial cycle. Periods of strong dollar risk appetite now lift both Bitcoin and U.S. equities, while periods of tight dollar conditions and low risk appetite depress them jointly. The rise and persistence of the post-2020 slope therefore mark a structural transition: Bitcoin has shifted from an asset that once moved largely independently of equities to one that behaves as a systematic, equity-like exposure linked to the broader dollar-centered global cycle.

The broader cross asset evidence for Ethereum, Litecoin, and Ripple in Tables 3–10 reinforces this conclusion.

Table 3: Post-2020 United States: Ethereum–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{ETH}_{USD})$	0.005 (0.003)	0.004 (0.003)	0.003 (0.003)	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.001* (0.001)	-0.001** (0.001)	-0.001** (0.001)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10Y \text{ Yield}$					0.025*** (0.004)	0.025*** (0.004)
$\Delta 1M \text{ Yield}$						0.003 (0.005)
Observations	1092	1092	1092	1092	1092	1092
Adj. R^2	0.001	0.064	0.074	0.667	0.679	0.679

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The Ethereum regressions display a pattern similar to Bitcoin. In the pre-2020 period (Table 3), the bivariate slope on $\Delta \log(\text{ETH}_{USD})$ is small and imprecisely estimated (0.005 in column (1)). As controls for common macro-financial shocks are introduced, the coefficient steadily declines and becomes weakly negative (around -0.004 in columns (4) to (6)), while the adjusted R^2 rises from 0.001 to roughly 0.68. This behaviour is consistent with a low-comovement environment: once broad macro factors such as commodities, the dollar, volatility, and interest rates are absorbed, Ethereum contributes little incremental explanatory power for U.S. equities. In terms of the model, this reflects a sizable parity wedge. Ethereum’s limited integration into dollar-centred portfolios prevented the token from loading on the U.S. pricing kernel, leaving equity returns largely unaffected by crypto movements.

Table 4: Post-2020 United States: Ethereum–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{ETH}_{USD})$	0.106*** (0.005)	0.099*** (0.005)	0.090*** (0.005)	0.029*** (0.004)	0.029*** (0.004)	0.029*** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.000*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.007*** (0.001)	-0.004*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10Y \text{ Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1M \text{ Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.158	0.189	0.223	0.647	0.648	0.648

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

After 2020 (Table 4), the Ethereum slope rises sharply. The bivariate coefficient is large and highly significant (0.106 in column (1)), and although reduced by the addition of controls, it remains economically meaningful and statistically robust at about 0.029 in columns (4) to (6). The adjusted R^2 increases from 0.16 to approximately 0.65. This pattern aligns with the model’s prediction that crypto–FX–equity linkages strengthen as the residual wedge narrows. The square-root variance ratio corroborates this interpretation: values remain close to unity in both periods ($\sqrt{\rho} \approx 1.006$ pre-2020; ≈ 0.986 post-2020), but the slightly lower

post-2020 figure signals tighter parity and a reduced residual gap. The post-2020 results therefore suggest that Ethereum has become more tightly connected to the global dollar asset environment, moving alongside shifts in global risk appetite and dollar funding conditions. As institutional participation expanded and liquidity deepened after 2020, Ethereum increasingly loaded on the same pricing kernel that drives U.S. equity returns, generating a sustained pattern of positive comovement.

Litecoin exhibits a similar transition. Before 2020 (Table 5), the slope on $\Delta \log(\text{LTC}_{USD})$ small and statistically fragile in the bivariate regression (0.005 in column (1)) and falls to values indistinguishable from zero once controls for commodities, the dollar, volatility, and yields are added (columns (4) to (6)). Meanwhile, the adjusted R^2 rises sharply from essentially zero to roughly 0.75 as these common shocks are absorbed, reflecting the dominant role of broad macro drivers in explaining equity variation. These results indicate that prior to 2020, Litecoin—like other thinly traded tokens—conveyed little systematic information for U.S. equities beyond standard macro factors. With shallow liquidity and fragmented trading venues, Litecoin did not consistently load on the U.S. pricing kernel, leaving a sizable parity wedge and a regime characterized by weak or hedge-like comovement.

Table 5: Pre-2020 United States: Litecoin–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{LTC}_{USD})$	0.005 (0.006)	0.005 (0.005)	0.004 (0.005)	-0.000 (0.003)	-0.000 (0.003)	-0.001 (0.003)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
ΔDollar			-0.006*** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.001** (0.001)
ΔVIX				-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
$\Delta 10Y \text{ Yield}$					0.024*** (0.005)	0.024*** (0.005)
$\Delta 1M \text{ Yield}$						0.003 (0.006)
Observations	691	691	691	691	691	691
Adj. R^2	-0.000	0.089	0.111	0.742	0.750	0.750

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Post-2020 (Table 6), by contrast, the Litecoin slope increases substantially. The bivariate coefficient becomes large and highly significant (0.074 in column (1)) and although attenuated by controls, it remains positive and economically meaningful at approximately 0.015 in columns (4) to (6). The adjusted R^2 increases from about 0.10 to nearly 0.64, and the square root variance ratio moves slightly closer to unity, indicating a smaller residual wedge in line with the model’s prediction.

This shift mirrors the broader integration of crypto assets into the global dollar cycle: Litecoin increasingly co-moves with U.S. equities when global risk appetite strengthens and declines jointly when dollar conditions tighten. In this regime, Litecoin functions as a risky dollar-denominated asset, transmitting rather than hedging equity-linked dollar risk. Its post-2020 behavior thus reflects a compressed parity wedge and a structural alignment of crypto returns with the same pricing kernel that governs U.S. equity markets.

Table 6: Post-2020 United States: Litecoin–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{LTC}_{USD})$	0.074*** (0.005)	0.068*** (0.005)	0.062*** (0.005)	0.015*** (0.003)	0.015*** (0.003)	0.015*** (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.007*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1\text{M Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.096	0.136	0.180	0.640	0.642	0.642

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Tether provides an important benchmark because it is explicitly designed to track the U.S. dollar. In the pre-2020 period (Table 7), the slope on $\Delta \log(\text{USDT}_{USD})$ is small and statistically insignificant across all specifications, even though the adjusted R^2 rises sharply when standard controls are added. This behaviour is consistent with the nature of a pegged asset: once the underlying dollar dynamics are accounted for, virtually no independent variation remains in USDT returns to explain equity movements. The relatively large square-root variance ratio observed before 2020 (≈ 1.100) likely reflects peg frictions or microstructure noise in secondary trading rather than a genuine equity risk exposure.

Table 7: Pre-2020 United States: Tether–Equity Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{USDT}_{USD})$	0.009 (0.028)	0.018 (0.027)	0.016 (0.027)	0.003 (0.016)	0.005 (0.016)	0.004 (0.016)
$\Delta \text{Commodity}$		0.004*** (0.001)	0.004*** (0.001)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.004*** (0.001)	-0.002** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.026*** (0.005)	0.026*** (0.005)
$\Delta 1\text{M Yield}$						0.003 (0.006)
Observations	879	879	879	879	879	879
Adj. R^2	-0.001	0.076	0.092	0.677	0.688	0.688

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Post-2020 United States Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{USDT}_{USD})$	-0.216 (0.237)	-0.159 (0.230)	-0.114 (0.223)	-0.022 (0.143)	-0.031 (0.143)	-0.030 (0.143)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1\text{M Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	-0.000	0.056	0.115	0.637	0.638	0.638

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The period after 2020 (Table 8) shows the same pattern. The bivariate slope becomes imprecisely negative and then converges to a small value near zero once controls are introduced, while the adjusted R^2 rises to about 0.64. The square root variance ratio remains elevated (≈ 6.586), again reflecting deviations associated with microstructure frictions in maintaining the peg. Unlike volatile crypto assets such as Bitcoin, Ethereum, or Litecoin, Tether does not load on the U.S. pricing kernel and exhibits no meaningful comovement with U.S. equities. This contrast reinforces the model's implication that only risk-bearing USD-denominated crypto assets transmit global dollar risk into equity markets, while pegged stablecoins, like Tether, remain mechanically anchored and do not participate in the global dollar cycle.

Ripple displays a broadly similar but more muted evolution in its linkage with U.S. equity returns. In the pre-2020 sample (Table 9), the bivariate elasticity of U.S. equities with respect to $\Delta \log(\text{XRP}_{USD})$ is small and statistically weak (0.006 in column (1)). Once common global drivers are introduced, the XRP coefficient rapidly attenuates toward zero. Over the same sequence of specifications, the adjusted R^2 increases dramatically, reaching roughly 0.75, indicating that these macro-financial variables absorb nearly all systematic variation in U.S. equity returns. In terms of the model, this reflects a sizable parity wedge and limited integration into dollar-centred portfolios, leaving Ripple largely detached from the U.S. pricing kernel prior to 2020.

Table 9: Pre-2020 United States Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{XRP}_{USD})$	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)	-0.000 (0.003)	-0.000 (0.003)	-0.000 (0.003)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
ΔDollar			-0.006*** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.001** (0.001)
ΔVIX				-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
$\Delta 10\text{Y Yield}$					0.024*** (0.005)	0.024*** (0.005)
$\Delta 1\text{M Yield}$						0.003 (0.006)
Observations	691	691	691	691	691	691
Adj. R^2	0.000	0.089	0.111	0.742	0.750	0.750

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 regressions (Table 10) show a distinct shift. Ripple exhibits a sizable and highly significant bivariate slope (0.049 in column (1)), and although this sensitivity diminishes once controls are included, it remains consistently positive and statistically robust at approximately 0.013 in columns (4)–(6). The adjusted R^2 rises from about 0.053 to ~ 0.642 . These changes align closely with the model’s prediction that, as the idiosyncratic wedge shrinks, the FX-adjusted crypto leg increasingly covaries with the same systematic shocks that drive equity markets. The square-root variance ratio ($\sqrt{\rho} \approx 1.003$ pre-2020; ≈ 0.996 post-2020) confirms this view: its slight movement toward unity is consistent with a reduced residual wedge and a tighter parity condition.

Together, these findings imply that Ripple, like Bitcoin and Ethereum, has become increasingly tied to the global dollar risk environment. As institutional participation and liquidity have grown, Ripple now reacts more directly to global swings in dollar risk appetite, behaving as an additional risky USD exposure that comoves with U.S. equities.

Table 10: Post-2020 United States Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{XRP}_{USD})$	0.049*** (0.005)	0.044*** (0.004)	0.041*** (0.004)	0.013*** (0.003)	0.013*** (0.003)	0.013*** (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.008*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10Y \text{ Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1M \text{ Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.053	0.099	0.152	0.640	0.642	0.642

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To summarize, in the United States, the pre-2020 estimates show that once standard global controls (commodities, dollar, volatility, and yields) are included, Bitcoin, Ethereum, Litecoin, and Ripple all contribute little to no incremental explanatory power for U.S. equity returns: their coefficients are small, often switch sign, and are not robust, while the fit of the regressions rises sharply as common shocks are absorbed. In the post-2020 period, this pattern reverses for the major volatile tokens. The Bitcoin slope is large in the bivariate specification and remains economically and statistically significant after controls (around 0.036), with similar behavior for Ethereum (around 0.029), Litecoin (around 0.015), and Ripple (around 0.013). By contrast, Tether’s coefficients remain indistinguishable from zero throughout. Together, these results indicate that, after 2020, non-pegged U.S. dollar crypto assets load systematically on U.S. equities in an equity-like manner, while a stablecoin does not, providing clear evidence of a strong crypto–FX–equity linkage in the U.S. data.

4.5 United Kingdom

In this subsection, we study whether USD-denominated crypto assets, once converted into pounds, move systematically with United Kingdom equity markets. Tables 11–20 report daily regressions of U.K. equity returns on GBP denominated crypto returns for the pre 2020 and post 2020 samples, progressively adding controls for commodities, the dollar, implied volatility, and the yield curve including term spreads.

Table 11: Pre-2020 United Kingdom Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{GBP})$	0.003 (0.002)	0.002 (0.002)	0.003 (0.002)	0.000 (0.002)	0.000 (0.002)	0.000 (0.002)	0.000 (0.002)	0.000 (0.002)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.005*** (0.000)	-0.003*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.025*** (0.003)	0.025*** (0.003)	0.022*** (0.003)	0.022*** (0.003)
$\Delta 1\text{M Yield}$						0.010* (0.006)	0.010* (0.006)	0.010* (0.006)
$\Delta 10\text{Y Spread}$							0.007** (0.003)	0.007** (0.003)
$\Delta 3\text{M Spread}$								-0.009* (0.005)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.137	0.160	0.307	0.322	0.322	0.323	0.323

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

For Bitcoin, the pre 2020 regressions in Table 11 show that the bivariate slope on $\Delta \log(\text{BTC}_{GBP})$ is small and statistically weak (0.003 in column (1)). As controls for common shocks are added, this coefficient shrinks toward zero and becomes economically negligible (approximately 0.000 in columns (4) to (8)), while the adjusted R^2 rises from essentially zero to about 0.32. This pattern indicates that pre-2020 Bitcoin movements contain little information about U.K. equities beyond what is already explained by global macro-financial factors. In the context of the model, the crypto-FX-equity wedge remains large in this period: Bitcoin trading was thinner, markets were more segmented, and the asset did not load tightly on the U.K. stochastic discount factor. Any apparent co-movement is largely accounted for by exposures to global shocks rather than by an intrinsic linkage between Bitcoin and U.K. equities.

Table 12: Post-2020 United Kingdom Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{GBP})$	0.059*** (0.006)	0.051*** (0.006)	0.045*** (0.006)	0.018*** (0.006)	0.016*** (0.006)	0.016*** (0.006)	0.016*** (0.006)	0.016*** (0.006)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.030*** (0.003)	0.030*** (0.003)	0.036*** (0.004)	0.036*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.042	0.098	0.189	0.272	0.302	0.303	0.308	0.308

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

In sharp contrast, the post-2020 regressions in Table 12 show a pronounced increase in the strength and stability of the crypto–equity relationship. In the bivariate specification, the slope on $\Delta \log(\text{BTC}_{GBP})$ is sizable and highly significant (0.059 in column (1)). Adding controls attenuates the coefficient but leaves it economically meaningful and robust, with estimates around 0.016 in columns (5) to (8). Over the same sequence of specifications, the adjusted R^2 increases from about 0.04 to roughly 0.31. This indicates that, in the post-2020 period, Bitcoin behaves as a systematic risk factor for the U.K. market: conditional on global conditions, positive Bitcoin shocks are associated with higher equity returns in sterling terms.

Although the square-root variance ratio $\sqrt{\rho}$ remains close to one in both subsamples ($\sqrt{\rho} \approx 1.006$ pre-2020 and ≈ 1.017 post-2020), indicating comparable residual parity, the slope dynamics reveal a structural shift from low or hedge-like behavior toward persistent positive comovement. Interpreted through the lens of the global financial cycle, Bitcoin appears increasingly integrated into the dollar-centric risk environment that shapes U.K. asset prices, with the GBP-converted crypto leg acting as an additional conduit for USD-denominated risk.

The Ethereum results exhibit a distinct but informative pattern. Pre-2020 (Table 13), the bivariate ETH coefficient is small but statistically significant (0.009 in column (1)); although attenuated, the slope remains modestly positive after controls (0.005 in columns (7)–(8)). The adjusted R^2 increases steadily from 0.007 to 0.187 across specifications. Unlike Bitcoin, therefore, early Ethereum returns retain a small but persistent equity-like signal even after absorbing macro-financial factors, though the magnitude is economically limited. This aligns with a regime where Ethereum’s wedge is smaller than Bitcoin’s—consistent with ETH’s earlier integration into decentralized finance (DeFi) and its comparatively higher correlation with speculative technology assets.

Table 13: Pre-2020 United Kingdom Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{GBP})$	0.009*** (0.003)	0.008*** (0.003)	0.008*** (0.003)	0.006** (0.003)	0.006** (0.003)	0.006** (0.003)	0.005* (0.003)	0.005* (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			0.001 (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.015*** (0.005)	0.014*** (0.005)	0.024*** (0.006)	0.024*** (0.006)
$\Delta 1\text{M Yield}$						0.015** (0.007)	0.016** (0.007)	0.016** (0.007)
$\Delta 10\text{Y Spread}$							-0.019*** (0.005)	-0.019*** (0.005)
$\Delta 3\text{M Spread}$								-0.010* (0.006)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.007	0.082	0.082	0.168	0.173	0.176	0.186	0.187

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 14: Post-2020 United Kingdom Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{GBP})$	0.048*** (0.005)	0.041*** (0.005)	0.035*** (0.004)	0.013*** (0.004)	0.012*** (0.004)	0.012*** (0.004)	0.013*** (0.004)	0.013*** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.031*** (0.003)	0.030*** (0.003)	0.037*** (0.004)	0.037*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.048	0.103	0.190	0.272	0.302	0.304	0.308	0.308

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Post-2020 (Table 14), however, the crypto–equity linkage strengthens considerably. The bivariate slope rises to 0.048, and even after introducing controls, ETH maintains a meaningful loading of 0.012–0.013 in columns (4) to (8). The adjusted R^2 grows from 0.048 to about 0.308. The near-unity values of $\sqrt{\rho}$ in both windows ($\sqrt{\rho} \approx 0.999$ pre 2020 and ≈ 1.006 post 2020) further suggest that while the residual structure of the parity wedge does not change dramatically, the economic relevance of the ETH signal does. This transition is consistent with the model’s prediction: as markets deepen and the wedge compresses, the GBP-adjusted ETH leg loads more strongly on the U.K. pricing kernel, reflecting ETH’s increasing role as a speculative, equity-sensitive global asset.

The Litecoin estimates are more muted. Pre-2020 (Table 15), the bivariate coefficient is small and only marginally significant (0.008 in column (1)), and after controls the slope converges toward approximately 0.006 and is statistically insignificant. The adjusted R^2 increases from 0.003 to about 0.236, indicating that global factors, not Litecoin, explain most U.K. equity variation in this period.

Table 15: Pre-2020 United Kingdom Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{GBP})$	0.008* (0.005)	0.008* (0.004)	0.009* (0.004)	0.007 (0.004)	0.007 (0.004)	0.006 (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta \text{Commodity}$		0.005*** (0.000)	0.004*** (0.001)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014* (0.007)	0.022*** (0.008)	0.022*** (0.008)
$\Delta 1\text{M Yield}$						0.023** (0.009)	0.023*** (0.009)	0.023*** (0.009)
$\Delta 10\text{Y Spread}$							-0.017** (0.007)	-0.017** (0.007)
$\Delta 3\text{M Spread}$								0.005 (0.010)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.003	0.122	0.121	0.220	0.224	0.230	0.236	0.236

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 16: Post-2020 United Kingdom Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{GBP})$	0.031*** (0.004)	0.026*** (0.004)	0.023*** (0.004)	0.006 (0.004)	0.006 (0.004)	0.006 (0.004)	0.006* (0.004)	0.006* (0.004)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.031*** (0.003)	0.030*** (0.003)	0.037*** (0.004)	0.037*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.025	0.086	0.179	0.270	0.300	0.302	0.306	0.306

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Post-2020 (Table 16), the bivariate LTC coefficient is noticeably larger (0.031 in column (1)), but once controls enter, the coefficient again settles near 0.006, albeit now marginally significant. Adjusted R^2 rises from 0.025 to approximately 0.31. The square-root variance ratios remain close to one in both periods ($\sqrt{\rho} \approx 1.000$ pre-2020; ≈ 1.008 post-2020) indicating similar residual wedge dispersion. Overall, Litecoin appears to transmit global conditions but carries limited independent systematic information for U.K. equities. The shift after 2020 is therefore reflected more in statistical significance, which is consistent with tighter crypto market integration, than in the magnitude of the fully-controlled beta.

The Tether results highlight the distinct behavior of a USD pegged stablecoin in a GBP setting. Pre-2020

(Table 17), the USDT coefficient is small but statistically positive (0.037 in column (1)), increasing slightly with controls (to about 0.046). Once the dollar index, volatility, and yields are included, however, it is clear that these global factors mostly drive U.K. equity variation. The adjusted R^2 increases from 0.002 to 0.192, consistent with the notion that, absent peg deviations, USDT carries minimal independent information.

Table 17: Pre-2020 United Kingdom Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{GBP})$	0.037* (0.021)	0.057*** (0.020)	0.059*** (0.021)	0.056*** (0.020)	0.057*** (0.020)	0.056*** (0.020)	0.046** (0.020)	0.046** (0.020)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.013** (0.006)	0.012* (0.006)	0.018*** (0.007)	0.019*** (0.007)
$\Delta 1\text{M Yield}$						0.014* (0.008)	0.014* (0.008)	0.014* (0.008)
$\Delta 10\text{Y Spread}$							-0.012** (0.006)	-0.012** (0.006)
$\Delta 3\text{M Spread}$								-0.011 (0.007)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.002	0.097	0.096	0.183	0.186	0.188	0.191	0.192

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 18: Post-2020 United Kingdom Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{GBP})$	-0.302*** (0.039)	-0.184*** (0.040)	0.312*** (0.048)	0.346*** (0.045)	0.282*** (0.045)	0.280*** (0.045)	0.272*** (0.045)	0.272*** (0.045)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.027*** (0.003)	0.027*** (0.003)	0.033*** (0.004)	0.033*** (0.004)
$\Delta 1\text{M Yield}$						0.007** (0.003)	0.007** (0.003)	0.007** (0.003)
$\Delta 10\text{Y Spread}$							-0.011*** (0.003)	-0.011*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.027	0.077	0.182	0.289	0.312	0.314	0.317	0.317

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Post-2020 (Table 18), the raw bivariate coefficient turns sharply negative (-0.302), reflecting episodes of temporary peg dislocations or safe-haven flows. After absorbing the standard global controls, the coefficient flips and becomes strongly positive (approximately 0.27 in columns (7)–(8)), with adjusted R^2 reaching

roughly 0.317. This pattern is consistent with the model: once common dollar-linked shocks are removed, residual USDT/GBP variation loads on the same equity-pricing kernel as other USD assets through the FX channel. The square root variance ratio is modest pre 2020 and significantly larger post 2020 ($\sqrt{\rho} \approx 1.126$ versus ≈ 2.171). The elevated post-2020 square-root variance ratio highlights that peg frictions create dispersion in sterling terms, yet the fully controlled slope still indicates a positive conditional relationship after 2020.

Ripple displays a somewhat different pattern. In the pre 2020 period (Table 19), the slope on $\Delta \log(\text{XRP}_{GBP})$ is small but consistently positive, moving from 0.010 in the bivariate specification to about 0.008 in the fully controlled regressions (columns (7) and (8)), while the adjusted R^2 rises from 0.004 to roughly 0.24. This indicates a modest equity related component that persists after controlling for standard macro factors, though again the magnitude is small relative to the contribution of the controls.

Table 19: Pre-2020 United Kingdom Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{GBP})$	0.010** (0.005)	0.011** (0.005)	0.011** (0.005)	0.009** (0.005)	0.009* (0.004)	0.009* (0.004)	0.008* (0.004)	0.008* (0.004)
$\Delta \text{Commodity}$		0.005*** (0.000)	0.004*** (0.001)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014* (0.007)	0.022*** (0.008)	0.022*** (0.008)
$\Delta 1\text{M Yield}$						0.023** (0.009)	0.023*** (0.009)	0.023*** (0.009)
$\Delta 10\text{Y Spread}$							-0.016** (0.007)	-0.016** (0.007)
$\Delta 3\text{M Spread}$								0.006 (0.010)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.004	0.124	0.123	0.221	0.226	0.232	0.237	0.237

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post 2020 period (Table 20), the bivariate XRP coefficient increases to 0.017 in column (1) but collapses to an economically negligible and statistically insignificant value of about 0.003 once the full set of controls is included (columns (4) to (8)), even as the adjusted R^2 improves from 0.009 to just above 0.30. Both periods exhibit square root variance ratios close to one ($\sqrt{\rho} \approx 0.999$ pre 2020 and ≈ 1.008 post 2020), implying similar residual parity. Together, these results suggest that any apparent strengthening of XRP-equity comovement in the raw post 2020 data is largely absorbed by common shocks; after conditioning on the global financial cycle variables, Ripple's incremental loading on U.K. equities is small and not robust.

Table 20: Post-2020 United Kingdom Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{GBP})$	0.017*** (0.004)	0.014*** (0.004)	0.013*** (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.031*** (0.003)	0.030*** (0.003)	0.037*** (0.004)	0.037*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.009	0.073	0.171	0.269	0.299	0.301	0.306	0.305

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To summarize, in the United Kingdom, the contrast between the two windows is straightforward: post-2020 coefficients are systematically larger and more equity-like than pre-2020. Before 2020, Bitcoin’s fully controlled slope is essentially zero and the other tokens (Ethereum, Litecoin, Ripple, Tether) generate only small positive loadings once macro controls are included, with most equity variation explained by commodities, the dollar, volatility, and yield curve movements. After 2020, Bitcoin and Ethereum both exhibit robust positive coefficients that remain statistically significant in the fully controlled specifications, indicating a stronger systematic link between GBP converted crypto returns and U.K. equities. Litecoin’s controlled beta remains modest but becomes more precisely estimated, Tether’s slope turns strongly positive once dollar and rate effects are netted out, consistent with its USD proxy role operating through the FX channel, and Ripple’s contribution largely disappears once common shocks are absorbed. Overall, the U.K. evidence points to a post 2020 environment in which major USD linked crypto assets, when viewed in pounds, participate more fully in the global dollar driven pricing kernel, strengthening the crypto equity linkage relative to the weak and fragmentary patterns observed in the pre 2020 period.

4.6 Eurozone

Next, we examine whether USD-denominated crypto assets, once converted into euros, exhibit systematic comovement with Eurozone equity markets.

Tables 21–30 report daily regressions of Eurozone equity returns on EUR-denominated crypto returns for the pre-2020 and post-2020 periods, adding controls for commodities, the broad dollar, implied volatility, and the euro-area yield curve including term spreads.

For Bitcoin, the pre-2020 estimates in Table 21 indicate very limited explanatory power. The bivariate slope on $\Delta \log(\text{BTC}_{EUR})$ is small and statistically weak (0.003 in column (1)), and once controls for global and euro-area macro-financial conditions are included, the coefficient shrinks to values economically indistinguishable from zero (between 0 and -0.001 across columns (4)–(8)). Over the same sequence of specifications, the adjusted R^2 rises from essentially zero to roughly 0.36. This pattern mirrors the U.K. and U.S. pre-2020 results and is consistent with a low-comovement or hedge-like regime: Eurozone equities were almost entirely driven by standard global factors—commodity prices, dollar movements, volatility shocks, and interest-rate dynamics—leaving little systematic role for Bitcoin after conditioning on these drivers. Within the model’s framework, this corresponds to a period with a large crypto-FX-equity wedge: fragmented market depth and weak integration prevented the Bitcoin leg from loading on the Eurozone stochastic discount factor.

Table 21: Pre-2020 Eurozone Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{EUR})$	0.003 (0.003)	0.002 (0.002)	0.003 (0.002)	-0.000 (0.002)	-0.000 (0.002)	-0.000 (0.002)	-0.001 (0.002)	-0.001 (0.002)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
ΔVIX				-0.003*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10Y \text{ Yield}$					0.032*** (0.003)	0.031*** (0.003)	0.008** (0.004)	0.007** (0.004)
$\Delta 1M \text{ Yield}$						0.015** (0.006)	0.013** (0.006)	0.013** (0.006)
$\Delta 10Y \text{ Spread}$							0.048*** (0.004)	0.048*** (0.004)
$\Delta 3M \text{ Spread}$								-0.015** (0.006)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.111	0.129	0.315	0.336	0.337	0.363	0.364

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 22: Post-2020 Eurozone Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{EUR})$	0.072*** (0.006)	0.064*** (0.006)	0.055*** (0.006)	0.023*** (0.006)	0.022*** (0.005)	0.022*** (0.005)	0.022*** (0.005)	0.022*** (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.010*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10Y \text{ Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.020*** (0.004)	0.021*** (0.004)
$\Delta 1M \text{ Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008*** (0.003)
$\Delta 10Y \text{ Spread}$							0.002 (0.005)	0.000 (0.005)
$\Delta 3M \text{ Spread}$								0.013* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.061	0.107	0.239	0.342	0.356	0.358	0.358	0.359

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-2020 results in Table 22 reveal a notable shift. The bivariate coefficient rises sharply to 0.072 (column (1)), indicating a sizeable unconditional association between Bitcoin returns (converted into euros) and Eurozone equity performance. Although the slope declines once common shocks are controlled for, it remains economically and statistically meaningful at around 0.022 in columns (4)–(8). The adjusted R^2 similarly increases from 0.061 in the bivariate regression to approximately 0.36 with the full set of controls. These estimates imply that in the post-2020 period Bitcoin behaves as a non-trivial systematic factor for Eurozone equities: conditional on global conditions, positive Bitcoin shocks are associated with higher equity

returns across the monetary union.

The square-root variance ratio $\sqrt{\rho}$ remains close to unity in both periods (approximately 1.007 pre-2020 and 1.009 post-2020), indicating that residual parity is broadly similar across regimes. Thus, the main change after 2020 stems not from altered dispersion of residual shocks but from a meaningful rise in Bitcoin’s systematic loading. Interpreted through the lens of the model and the global financial cycle, this suggests that Bitcoin has become increasingly integrated into the broader dollar-centered risk environment that influences Eurozone asset prices. As crypto markets deepen, institutional participation expands, and cross-border trading infrastructure matures, Bitcoin begins to co-move more tightly with global risk factors—particularly those emanating from the U.S. dollar funding cycle. Consequently, Bitcoin’s EUR-denominated returns act as an additional conduit through which global risk appetite and dollar conditions propagate into Eurozone equities.

Turning to Ethereum, the pre 2020 results again reveal very limited explanatory power. Table 23 shows that the coefficient on $\Delta \log(\text{ETH}_{EUR})$ declines from a small positive 0.005 in the bivariate regression to an economically negligible value of approximately 0.001 in the fully controlled specifications (columns (4) to (8)). The adjusted R^2 increases from 0.002 to 0.232, which confirms that once global factors are accounted for, the remaining marginal effect of Ethereum on Eurozone equities is very small. This is consistent with a period in which the crypto market was still insufficiently integrated with dollar funding conditions and European risk taking, resulting in a sizeable wedge and little direct transmission.

Table 23: Pre-2020 Eurozone Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{EUR})$	0.005* (0.003)	0.005 (0.003)	0.005 (0.003)	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.000 (0.001)	0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10Y \text{ Yield}$					0.020*** (0.005)	0.020*** (0.005)	-0.001 (0.007)	-0.001 (0.007)
$\Delta 1M \text{ Yield}$						0.016** (0.006)	0.012** (0.006)	0.013** (0.006)
$\Delta 10Y \text{ Spread}$							0.040*** (0.008)	0.040*** (0.008)
$\Delta 3M \text{ Spread}$								0.008 (0.010)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.002	0.056	0.055	0.201	0.212	0.215	0.232	0.232

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 24: Post-2020 Eurozone Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{EUR})$	0.061*** (0.005)	0.054*** (0.005)	0.046*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.020*** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.010*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.004)	0.021*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008*** (0.003)
$\Delta 10\text{Y Spread}$							0.001 (0.005)	-0.001 (0.005)
$\Delta 3\text{M Spread}$								0.012* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.076	0.120	0.246	0.344	0.359	0.361	0.360	0.361

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-2020 results differ substantially. In the bivariate specification, the slope on $\Delta \log(\text{ETH}_{EUR})$ is large and strongly significant (0.061 in column (1)), and although the coefficient declines once controls are introduced, it remains economically meaningful at roughly 0.020 to 0.021 in columns (4) to (8). The adjusted R^2 simultaneously increases from 0.076 to about 0.361, reflecting a substantial improvement in explanatory power. These findings are consistent with a Eurozone pricing environment in which Ethereum becomes more tightly integrated into the global dollar centred risk cycle.

As market depth increases and institutional participation expands, the idiosyncratic wedge that separates crypto returns from traditional asset pricing dynamics compresses, allowing Ethereum to load more strongly on the same systematic forces that drive Eurozone equities. This systematic linkage is further supported by the behaviour of the square root variance ratio, which remains close to unity in both periods ($\sqrt{\rho} \approx 1.003$ before 2020 and ≈ 0.998 after 2020). The similarity of these ratios indicates that residual dispersion is nearly unchanged across windows, which implies that the shift in comovement is driven primarily by higher Ethereum loadings rather than changes in unexplained volatility. Overall, the post-2020 evidence points to a meaningful and persistent integration of Ethereum into the broader Eurozone risk environment, with positive Ethereum shocks now accompanied by contemporaneous increases in Eurozone equity returns under similar global financial conditions.

For Litecoin, the pre-2020 estimates indicate that once controls for broad macroeconomic conditions are introduced, the association between its returns and Eurozone equity returns becomes economically small and statistically modest. In the bivariate specification, the slope on $\Delta \log(\text{LTC}_{EUR})$ is slightly positive (0.012 in column (1)), and although it remains positive after adding successive controls, it converges to a very small value of approximately 0.010 in columns (4)–(8). Over the same sequence of specifications, the adjusted R^2 rises from 0.008 to about 0.298. This pattern suggests that before 2020, most of the variation in Eurozone equities was driven by common global forces rather than by movements in Litecoin. Within the conceptual framework of the model, this reflects a period in which the crypto FX equity wedge remained relatively large, so the Litecoin leg did not load tightly on the Eurozone pricing kernel and any apparent association between the two series was largely mediated by standard global factors.

Table 25: Pre-2020 Eurozone Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{EUR})$	0.012** (0.005)	0.012*** (0.004)	0.012*** (0.004)	0.010** (0.004)	0.010** (0.004)	0.009** (0.004)	0.010** (0.004)	0.010** (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.001)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014** (0.007)	-0.007 (0.009)	-0.007 (0.009)
$\Delta 1\text{M Yield}$						0.025*** (0.008)	0.022** (0.008)	0.022*** (0.008)
$\Delta 10\text{Y Spread}$							0.042*** (0.011)	0.042*** (0.011)
$\Delta 3\text{M Spread}$								0.017 (0.014)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.008	0.109	0.117	0.271	0.276	0.284	0.298	0.298

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 26: Post-2020 Eurozone Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{EUR})$	0.036*** (0.004)	0.031*** (0.004)	0.027*** (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.001** (0.000)	0.001*** (0.000)
ΔDollar			-0.011*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.004)	0.022*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008*** (0.003)
$\Delta 10\text{Y Spread}$							0.000 (0.005)	-0.001 (0.005)
$\Delta 3\text{M Spread}$								0.013* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.034	0.086	0.222	0.337	0.352	0.354	0.354	0.355

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 estimates tell a different story. The bivariate slope on $\Delta \log(\text{LTC}_{EUR})$ becomes larger and highly significant (0.036 in column (1)), consistent with a period in which Litecoin appreciates in episodes of stronger global risk appetite and dollar liquidity. After incorporating the full set of controls, the coefficient declines to a small but statistically significant value of approximately 0.007 in columns (4)–(8). At the same time, the adjusted R^2 improves from 0.034 to around 0.355. The fact that the fully controlled slope is modest in magnitude, yet persistent and precisely estimated, indicates that Litecoin has begun to embed a mild but systematic exposure to the Eurozone pricing kernel. This is consistent with increasing market depth and

greater integration of crypto assets into global dollar related portfolios after 2020.

The square root variance ratio provides additional insight into the changing structure of the parity relationship. Both the pre 2020 and post 2020 values lie very close to unity ($\sqrt{\rho} \approx 0.996$ before 2020 and $\sqrt{\rho} \approx 1.007$ after 2020), suggesting that the relative dispersion between the Litecoin leg and the benchmark equity return is similar across the two periods. This reinforces the conclusion that the main evolution over time arises from an improvement in explanatory power and statistical precision rather than from a fundamental change in the residual parity structure.

Overall, the Eurozone evidence for Litecoin suggests a shift from weak and somewhat incidental association before 2020 toward a more stable and persistent, although still moderate, equity linked exposure in the later sample window.

Tether displays a distinct pattern driven by its stablecoin nature. The pre-2020 estimates indicate that once controls for broad macroeconomic conditions are introduced, the association between Litecoin returns and Eurozone equity returns becomes economically small and statistically modest. In the bivariate specification, the slope on $\Delta \log(\text{LTC}_{EUR})$ is slightly positive (0.012 in column (1)), and although it remains positive after adding successive controls for commodities, the dollar, implied volatility, interest rates, and term spreads, it converges to a very small value of approximately 0.010 in columns (4)–(8). Over the same sequence of specifications, the adjusted R^2 rises from 0.008 to about 0.298. This pattern suggests that before 2020 most of the variation in Eurozone equities was driven by common global forces rather than by movements in Litecoin. Within the conceptual framework of the model, this reflects a period in which the crypto FX equity wedge remained relatively large, so the Litecoin leg did not load tightly on the Eurozone pricing kernel, and any apparent association between the two series was largely mediated by standard global factors.

Table 27: Pre-2020 Eurozone Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{EUR})$	0.002 (0.022)	0.019 (0.021)	0.026 (0.022)	0.016 (0.020)	0.016 (0.020)	0.015 (0.020)	0.025 (0.020)	0.023 (0.020)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.002* (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	0.000 (0.001)	0.000 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.018*** (0.006)	0.017*** (0.006)	-0.006 (0.008)	-0.006 (0.008)
$\Delta 1\text{M Yield}$						0.017** (0.007)	0.014* (0.007)	0.014* (0.007)
$\Delta 10\text{Y Spread}$							0.045*** (0.010)	0.045*** (0.010)
$\Delta 3\text{M Spread}$								0.012 (0.012)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	-0.001	0.072	0.074	0.210	0.217	0.221	0.239	0.239

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 28: Post-2020 Eurozone Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{EUR})$	-0.319*** (0.047)	-0.238*** (0.047)	0.605*** (0.057)	0.484*** (0.053)	0.438*** (0.053)	0.432*** (0.053)	0.451*** (0.054)	0.451*** (0.054)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.016*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.018*** (0.003)	0.018*** (0.003)	0.013*** (0.004)	0.014*** (0.004)
$\Delta 1\text{M Yield}$						0.007** (0.003)	0.007** (0.003)	0.007** (0.003)
$\Delta 10\text{Y Spread}$							0.008* (0.005)	0.006 (0.005)
$\Delta 3\text{M Spread}$								0.014** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.021	0.072	0.244	0.362	0.372	0.373	0.373	0.374

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 estimates tell a different story. The bivariate slope on $\Delta \log(\text{LTC}_{EUR})$ becomes larger and highly significant (0.036 in column (1)), consistent with a period in which Litecoin appreciates in episodes of stronger global risk appetite and dollar liquidity. After incorporating the full set of controls, the coefficient declines to a small but statistically significant value of approximately 0.007 in columns (4)–(8). At the same time, the adjusted R^2 improves from 0.034 to around 0.355. The fact that the fully controlled slope is modest in magnitude, yet persistent and precisely estimated, indicates that Litecoin has begun to embed a mild but systematic exposure to the Eurozone pricing kernel. This is consistent with increasing market depth and greater integration of crypto assets into global dollar related portfolios after 2020.

The square root variance ratio provides additional insight into the changing structure of the parity relationship. Both the pre 2020 and post 2020 values lie very close to unity ($\sqrt{\rho} \approx 0.996$ before 2020 and $\sqrt{\rho} \approx 1.007$ after 2020), suggesting that the relative dispersion between the Litecoin leg and the benchmark equity return is similar across the two periods. This reinforces the conclusion that the main evolution over time arises from an improvement in explanatory power and statistical precision rather than from a fundamental change in the residual parity structure.

Finally, for Ripple, once controls for broad macroeconomic forces are introduced, the return sensitivity of euro-denominated Ripple to Eurozone equities remains small but consistently positive. The bivariate specification produces a coefficient of 0.009 in column (1), and as successive controls are added, the slope stabilizes at approximately 0.007 to 0.008 in columns (4) to (8). Although these estimates are only marginally significant, the adjusted R^2 rises from 0.004 to 0.296, which confirms that the explanatory power in this period is driven largely by the macro factors themselves. Ripple, therefore, carries only a modest equity-like footprint before 2020, consistent with a setting in which crypto markets remain thin and segmented from the pricing kernel that governs Eurozone equity returns. In such an environment, the crypto–FX–equity wedge remains sizable. Crypto returns contain idiosyncratic movements unrelated to the broader global cycle, and much of the observed comovement vanishes once standard global controls are absorbed.

Table 29: Pre-2020 Eurozone Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{EUR})$	0.009* (0.005)	0.010** (0.005)	0.010** (0.005)	0.007* (0.004)	0.007* (0.004)	0.007 (0.004)	0.008* (0.004)	0.007* (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.001)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014** (0.007)	-0.007 (0.009)	-0.007 (0.009)
$\Delta 1\text{M Yield}$						0.025*** (0.008)	0.022*** (0.008)	0.023*** (0.008)
$\Delta 10\text{Y Spread}$							0.042*** (0.011)	0.042*** (0.011)
$\Delta 3\text{M Spread}$								0.017 (0.014)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.004	0.105	0.113	0.268	0.273	0.281	0.295	0.296

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 30: Post-2020 Eurozone Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{EUR})$	0.020*** (0.004)	0.017*** (0.004)	0.016*** (0.003)	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.011*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.004)	0.022*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							0.000 (0.005)	-0.001 (0.005)
$\Delta 3\text{M Spread}$								0.013* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.013	0.070	0.212	0.337	0.351	0.353	0.353	0.354

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-2020 patterns display a shift in the unconditional linkage but not in the fully controlled slope. The raw bivariate coefficient rises to 0.020 in column (1), indicating stronger unconditional comovement between Ripple and Eurozone equities in the later period. After accounting for the same set of global controls, however, the coefficient compresses sharply to values near 0.003 in columns (4) to (8), and the statistical significance largely disappears. At the same time, the adjusted R^2 again increases substantially, from 0.013 to about 0.354, reaffirming that macroeconomic variables remain the dominant determinants of Eurozone equity variation. These outcomes suggest that the increase in the unconditional Ripple–equity association after 2020

largely reflects shared exposure to common global shocks rather than a deeper structural integration of Ripple into the Eurozone pricing kernel. Once these common shocks are filtered out, little incremental information remains in $\Delta \log(\text{XRP}_{EUR})$.

The square root variance ratio provides additional insight. Both periods lie extremely close to unity, with values of $\sqrt{\rho} \approx 0.9987$ before 2020 and $\sqrt{\rho} \approx 1.0070$ after 2020. This pattern indicates that the residual parity wedge behaves similarly across periods and that the unexplained component of the crypto–equity linkage does not meaningfully change. The evolution over time is therefore not due to a shift in residual dispersion but rather reflects changes in the unconditional exposure of Ripple to the global environment. Overall, the Eurozone evidence confirms that Ripple maintains at best a weak and economically limited equity like loading, and that most of the observed correlation with equity returns is mediated through broad global financial conditions rather than intrinsic integration with local asset pricing forces.

To summarize, in the Eurozone, the main message is the strengthening of the linkage after 2020. In the pre-2020 sample, once we control for commodities, the dollar, volatility, rates, and term spreads, Bitcoin and Ethereum are essentially washed out, and the remaining tokens only leave small, fragile positive slopes—most of the action is common global factors rather than crypto-specific exposure. In the post-2020 sample, the coefficients on Bitcoin and Ethereum become clearly larger and remain statistically and economically meaningful even in the fully controlled regressions, Litecoin retains a modest positive loading, and Tether turns into a strong positive predictor once dollar and rate effects are netted out, consistent with a USD-proxy/FX channel. Lastly, Ripple’s incremental role fades once controls are included but remains positive. Overall, this shift from weak or incidental pre-2020 effects to stronger, persistent post-2020 betas shows that the crypto–FX–equity linkage becomes materially tighter for the Eurozone in the later period.

4.7 Japan

In this subsection, we study whether USD-denominated crypto assets, once converted into yen, move systematically with Japanese equity markets.

Tables 31–40 present the daily regressions of Japanese equity returns on the yen converted crypto legs for the pre 2020 and post 2020 subsamples, sequentially introducing controls for commodities, the broad dollar index, implied volatility, domestic and United States interest rates, and term structure variables. This specification allows us to trace how much explanatory power the crypto leg retains once the main global and domestic risk variables that typically drive Japanese equities are accounted for.

For Bitcoin, the pre-2020 results in Table 31 show that the bivariate slope on $\Delta \log(\text{BTC}_{JPY})$ is small and statistically weak (0.004 in column (1)). As the full set of controls is added, the coefficient diminishes toward zero and loses all economic relevance, stabilizing at approximately 0.000 in columns (4) to (8). During the same sequence, the adjusted R^2 rises notably from near zero to roughly 0.29. This pattern is characteristic of a low comovement regime in which broad macro forces, including movements in global commodities, the dollar, measures of volatility, and the yield curve, explain nearly all of the variation in Japanese equities. The Bitcoin leg therefore provides little incremental information prior to 2020. From the perspective of the model, this corresponds to a period in which the crypto FX equity wedge remains large. Bitcoin did not price off the same stochastic discount factor that governs Japanese equities, and any modest correlation that appeared in the raw regressions is absorbed once global financial conditions are controlled for.

Table 31: Pre-2020 Japan Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{JPY})$	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)	0.002 (0.003)	0.001 (0.003)	0.001 (0.003)	-0.001 (0.003)	-0.001 (0.003)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.001 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10Y \text{ Yield}$					0.012** (0.005)	0.012** (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta 1M \text{ Yield}$						-0.011 (0.010)	-0.012 (0.009)	-0.012 (0.009)
$\Delta 10Y \text{ Spread}$							0.163*** (0.012)	0.163*** (0.012)
$\Delta 3M \text{ Spread}$								0.019** (0.009)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.011	0.011	0.019	0.020	0.020	0.072	0.073

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 32: Post-2020 Japan Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{JPY})$	0.052*** (0.008)	0.045*** (0.008)	0.041*** (0.008)	0.033*** (0.008)	0.031*** (0.008)	0.031*** (0.008)	0.034*** (0.008)	0.034*** (0.008)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.009* (0.005)	0.009* (0.005)	0.004 (0.005)	0.004 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta 10Y \text{ Spread}$							0.129*** (0.013)	0.128*** (0.013)
$\Delta 3M \text{ Spread}$								0.009 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.020	0.033	0.050	0.053	0.055	0.054	0.094	0.094

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 Bitcoin regressions in Table 32 reveal a clear strengthening of the crypto equity linkage. In the bivariate specification, the coefficient rises sharply to 0.067 (column (1)), indicating that yen-converted Bitcoin returns began to move in a more equity-like fashion with Japanese stocks. Once controls are introduced, the coefficient declines but remains economically and statistically meaningful at approximately 0.015 across columns (4) to (8). The adjusted R^2 increases from about 0.05 to nearly 0.34. These results indicate that after 2020, Bitcoin became more tightly linked with the pricing kernel relevant for Japanese assets, consistent with the model's prediction that a reduction in the wedge variance binds the crypto leg more

strongly to global risk conditions. The square root variance ratio $\sqrt{\rho}$ remains close to unity in both periods ($\sqrt{\rho} \approx 1.004$ in the earlier window and ≈ 1.011 after 2020), suggesting similar residual parity across time. The key change is therefore the magnitude and robustness of the slope, not a shift in residual dispersion.

Turning to Ethereum, the pre 2020 regressions show an even weaker connection with Japanese equities. In the bivariate specification the slope on $\Delta \log(\text{ETH}_{JPY})$ is essentially zero (0.002 in column (1)), and once macro controls are introduced it converges entirely to a value that is economically negligible (approximately zero in columns (4) to (8)). The fit improves only slightly across specifications, with the adjusted R^2 moving from a marginally negative value to roughly 0.117. These patterns are consistent with Ethereum playing almost no independent role in pricing Japanese equities during the earlier period. Broader global and domestic drivers absorb virtually all systematic variation, and Ethereum does not appear to embed meaningful information about the local pricing kernel.

Table 33: Pre-2020 Japan Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{JPY})$	0.002 (0.004)	0.001 (0.004)	0.001 (0.004)	0.000 (0.004)	-0.000 (0.004)	-0.000 (0.004)	0.000 (0.004)	0.001 (0.004)
$\Delta \text{Commodity}$		0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ΔVIX				-0.000** (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000** (0.000)	-0.000** (0.000)
$\Delta 10Y \text{ Yield}$					0.024*** (0.008)	0.025*** (0.008)	0.010 (0.008)	0.010 (0.008)
$\Delta 1M \text{ Yield}$						-0.009 (0.010)	-0.006 (0.009)	-0.006 (0.009)
$\Delta 10Y \text{ Spread}$							0.244*** (0.024)	0.245*** (0.024)
$\Delta 3M \text{ Spread}$								0.021*** (0.008)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	-0.001	0.015	0.014	0.019	0.026	0.026	0.111	0.117

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-2020 regressions display a noticeable strengthening of the relationship between Ethereum returns and Japanese equity performance. The bivariate ETH coefficient is meaningfully larger and highly significant (0.045 in column (1)), and although the magnitude declines once controls are introduced, the coefficient remains economically relevant at approximately 0.030 in columns (4) to (8). The adjusted R^2 also improves, rising from about 0.027 in the bivariate model to roughly 0.096 in the full specification. These patterns indicate that after 2020, Ethereum shocks coincide more systematically with movements in Japanese equities even after taking account of global financial conditions. This is consistent with the idea that increased market depth, greater institutional participation, and tighter integration with dollar based funding markets compress the crypto to foreign exchange wedge, thereby aligning Ethereum more closely with the same risk environment that influences Japanese equities.

Table 34: Post-2020 Japan Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{JPY})$	0.045*** (0.006)	0.040*** (0.006)	0.036*** (0.006)	0.031*** (0.006)	0.030*** (0.006)	0.030*** (0.006)	0.030*** (0.006)	0.030*** (0.006)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000** (0.000)	-0.000** (0.000)	-0.000** (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.010** (0.005)	0.010** (0.005)	0.005 (0.005)	0.005 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.004)	0.001 (0.004)
$\Delta 10Y \text{ Spread}$							0.126*** (0.013)	0.126*** (0.013)
$\Delta 3M \text{ Spread}$								0.008 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.027	0.039	0.055	0.057	0.059	0.058	0.097	0.096

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The square root variance ratio provides a complementary diagnostic. Both windows produce values very near unity, with $\sqrt{\rho} \approx 1.008$ before 2020 and $\sqrt{\rho} \approx 1.010$ afterwards. This implies that residual dispersion in the parity condition is similarly small across the two periods. The improvement in explanatory power therefore reflects not changes in unexplained return variability but rather the emergence of a more stable and persistent Ethereum loading on Japanese equities in the latter period. In summary, the post2020 evidence points to a modest but economically meaningful incorporation of Ethereum into the broader global risk structure that affects Japanese equity returns.

Litecoin shows a more muted transition, but still displays evidence of a shift in its relationship with Japanese equities. In the pre 2020 sample (Table 35), once controls for common shocks are added, the Litecoin coefficient remains essentially indistinguishable from zero. In the bivariate specification, the slope on $\Delta \log(\text{LTC}_{JPY})$ is slightly negative (-0.002 in column (1)), and as controls for commodities, the dollar, implied volatility, and interest rates are introduced, the coefficient stays near zero (ranging from -0.003 in columns (4) to (7) and settling at -0.001 in column (8)). Over this sequence of specifications, the adjusted R^2 rises from a negligible value of -0.001 to approximately 0.158

Table 35: Pre-2020 Japan Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{JPY})$	-0.002 (0.006)	-0.002 (0.006)	-0.002 (0.006)	-0.003 (0.006)	-0.003 (0.006)	-0.003 (0.006)	-0.003 (0.006)	-0.001 (0.006)
$\Delta \text{Commodity}$		0.003*** (0.001)	0.002*** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
ΔDollar			-0.004*** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.004*** (0.001)	-0.004** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.026** (0.010)	0.027** (0.011)	0.008 (0.010)	0.008 (0.010)
$\Delta 1M \text{ Yield}$						-0.017 (0.013)	-0.013 (0.012)	-0.012 (0.012)
$\Delta 10Y \text{ Spread}$							0.263*** (0.028)	0.265*** (0.028)
$\Delta 3M \text{ Spread}$								0.026*** (0.009)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.020	0.029	0.030	0.037	0.038	0.148	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 period, the Litecoin slope becomes positive and statistically significant in the bivariate specification (0.019 in column (1)), indicating that raw comovement with Japanese equities strengthened after 2020. However, once the full set of controls is introduced, the coefficient is largely absorbed by the same global shocks. After conditioning on commodities, the dollar, volatility, and the yield curve, the Litecoin loading compresses to approximately 0.006 and loses statistical significance (columns (4) to (8)), while the adjusted R^2 rises from 0.006 to nearly 0.087. This indicates that, although Litecoin displays a stronger unconditional relation with Japanese equities after 2020, the systematic component of that relation remains small once global conditions are taken into account. The shift is therefore driven primarily by the increased explanatory power of global macro variables in the Japanese equity environment, rather than by a markedly higher fully controlled Litecoin beta.

The square root variance ratio $\sqrt{\rho}$ offers complementary insight. In both subsamples the ratio lies close to unity ($\sqrt{\rho} \approx 1.013$ before 2020 and $\sqrt{\rho} \approx 1.024$ after 2020), which indicates that the residual parity wedge remains small relative to the idiosyncratic component of the crypto return. The similarity across periods implies that the structural tightness of the parity condition is broadly unchanged. Consequently, the main difference between the two windows arises from the behavior of the estimated slope: although raw comovement strengthens in the later period, the fully controlled equity-like loading remains modest. This is consistent with a scenario in which Litecoin becomes somewhat more influenced by global financial conditions after 2020 but still does not transmit those conditions into the Japanese equity market in a significant way once the major macro channels are accounted for.

Table 36: Post-2020 Japan Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{JPY})$	0.019*** (0.005)	0.015*** (0.005)	0.013** (0.005)	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)	0.006 (0.005)	0.006 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000** (0.000)	-0.000** (0.000)
$\Delta 10Y \text{ Yield}$					0.011** (0.005)	0.011** (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta 10Y \text{ Spread}$							0.127*** (0.013)	0.127*** (0.013)
$\Delta 3M \text{ Spread}$								0.008 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.006	0.021	0.040	0.047	0.049	0.048	0.087	0.087

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Ripple exhibits a similar structure. Table 39 shows that prior to 2020, the bivariate specification produces a coefficient of roughly 0.030 in column (1), but once controls for commodities, the dollar, volatility, interest rates, and term spreads are included, the estimates settle in the narrow range of approximately 0.019 to 0.027 across columns (5) to (8), with none of these coefficients reaching conventional significance levels. The adjusted R^2 increases only modestly from essentially zero to about 0.130 as common shocks are absorbed, suggesting that in the period before 2020 the yen denominated stablecoin return contains little systematic information beyond that already captured by broad global risk factors. This behaviour is consistent with the fact that a USD pegged asset should have minimal independent variation once dollar related movements and global volatility conditions are controlled for. The large value of the square root variance ratio in this period ($\sqrt{\rho} \approx 1.295$) further indicates substantial residual dispersion, which is consistent with peg frictions, microstructure noise, and short lived deviations from one to one parity when measured in JPY terms.

Table 37: Pre-2020 Japan Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{JPY})$	0.030 (0.030)	0.034 (0.030)	0.037 (0.030)	0.030 (0.030)	0.019 (0.031)	0.020 (0.031)	0.027 (0.029)	0.027 (0.029)
$\Delta \text{Commodity}$		0.003*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.003** (0.001)	-0.003* (0.001)	-0.003** (0.001)	-0.003* (0.001)	-0.003** (0.001)	-0.003** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.023** (0.009)	0.024*** (0.009)	0.007 (0.009)	0.007 (0.009)
$\Delta 1M \text{ Yield}$						-0.015 (0.011)	-0.011 (0.011)	-0.010 (0.011)
$\Delta 10Y \text{ Spread}$							0.251*** (0.026)	0.252*** (0.026)
$\Delta 3M \text{ Spread}$								0.021*** (0.008)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	-0.000	0.021	0.025	0.026	0.032	0.033	0.125	0.130

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 38: Post-2020 Japan Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{JPY})$	0.148*** (0.048)	0.168*** (0.047)	0.363*** (0.051)	0.322*** (0.053)	0.331*** (0.059)	0.331*** (0.059)	0.377*** (0.058)	0.378*** (0.058)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
ΔVIX				-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					-0.002 (0.005)	-0.002 (0.005)	-0.009* (0.005)	-0.009* (0.005)
$\Delta 1M \text{ Yield}$						-0.001 (0.005)	0.001 (0.004)	0.001 (0.004)
$\Delta 10Y \text{ Spread}$							0.134*** (0.013)	0.134*** (0.013)
$\Delta 3M \text{ Spread}$								0.009 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.004	0.023	0.060	0.063	0.062	0.062	0.105	0.105

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In contrast, the post 2020 regressions reveal a very different pattern. The raw bivariate coefficient rises sharply to 0.148 in column (1), and after controlling for the full set of macro financial variables, the slope remains substantially large at approximately 0.32 to 0.38 across columns (4) to (8). While the adjusted R^2 increases only modestly from 0.004 to about 0.105, the persistence of these large coefficients points to a meaningful association between USDT returns and Japanese equity returns that survives even after common risk drivers are absorbed. In a yen based setting, where global safe haven flows, carry conditions, and JPY/USD dynamics play central roles, the stablecoin behaves less as a pure peg and more as an instrument

transmitting dollar related shocks that are not fully captured by our standard controls. This may occur because yen strength or weakness interacts with dollar liquidity demand in a way that amplifies small deviations from the peg, thereby introducing co movement with domestic risky assets.

The square root variance ratio reinforces this interpretation: it rises substantially to $\sqrt{\rho} \approx 2.711$ in the post-2020 sample (from $\sqrt{\rho} \approx 1.295$ pre-2020), indicating pronounced residual volatility around the parity condition that is consistent with more active peg adjustments, liquidity effects, and JPY USD microstructure noise. Although the fully controlled coefficients are large, part of their magnitude likely reflects these residual frictions rather than pure exposure to equity related risk, making the interpretation of USDT in the Japanese context more complex than for other crypto assets.

Finally, the pre-2020 results show that the Ripple coefficient in Table 39 is small and statistically fragile at every step of the specification. The bivariate regression produces a modest and imprecise loading on $\Delta \log(\text{XRP}_{JPY})$ (0.006 in column (1)). Once the standard global controls are introduced, this coefficient declines toward an economically negligible value of approximately 0.003 in columns (7) and (8) and loses statistical significance. Over the same sequence of specifications, the adjusted R^2 rises from essentially zero to roughly 0.158, indicating that broad movements in commodities, the dollar, global volatility, and the Japanese term structure account for most of the variation in Japanese equity returns before 2020. Ripple offers little additional explanatory content once these common forces are absorbed, which is consistent with a period in which the parity wedge between the crypto leg and the local pricing kernel remains large and prevents Ripple from aligning with the dominant drivers of equity returns in Japan. The square root variance ratio supports this interpretation. The estimate $\sqrt{\rho} \approx 1.010$ lies very close to unity, which reflects a residual wedge that is small relative to the idiosyncratic crypto component but still sufficiently persistent to prevent a meaningful incremental Ripple loading.

Table 39: Pre-2020 Japan Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{JPY})$	0.006 (0.007)	0.006 (0.007)	0.005 (0.007)	0.005 (0.007)	0.004 (0.007)	0.004 (0.007)	0.003 (0.006)	0.003 (0.006)
$\Delta \text{Commodity}$		0.003*** (0.001)	0.002*** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
ΔDollar			-0.004*** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.004*** (0.001)	-0.004** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.025** (0.010)	0.027** (0.011)	0.008 (0.010)	0.008 (0.010)
$\Delta 1M \text{ Yield}$						-0.017 (0.013)	-0.013 (0.012)	-0.012 (0.012)
$\Delta 10Y \text{ Spread}$							0.263*** (0.028)	0.264*** (0.028)
$\Delta 3M \text{ Spread}$								0.026*** (0.009)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.000	0.021	0.029	0.031	0.037	0.038	0.148	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the period after 2020, Table 40 shows a similarly muted pattern. The bivariate slope on $\Delta \log(\text{XRP}_{JPY})$ is statistically significant but remains small in magnitude (0.011 in column (1)). Once the full set of global and domestic controls is included, the coefficient is absorbed almost entirely and stabilizes around 0.003 in columns (4) to (8), with no statistical significance. The adjusted R^2 improves only modestly from 0.002 to approximately 0.086, and virtually all of this improvement is driven by the macro controls rather than by Ripple. The dollar factor, the VIX, and long short yield spreads contribute strongly to explanatory power, indicating that the principal forces shaping Japanese equity returns in this period are global in nature. Ripple

does not transmit additional systematic information once those global forces are accounted for. The square root variance ratio for the post 2020 window, $\sqrt{\rho} \approx 1.019$, again lies very close to one, implying a residual parity structure that is largely unchanged across periods and reinforcing that the limited explanatory role of Ripple is not due to shifts in residual dispersion but rather to the absence of a meaningful fully controlled beta.

Table 40: Post-2020 Japan Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{JPY})$	0.011** (0.005)	0.008* (0.005)	0.007 (0.005)	0.003 (0.005)	0.003 (0.005)	0.003 (0.005)	0.003 (0.005)	0.003 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
$\Delta 10Y \text{ Yield}$					0.011** (0.005)	0.011** (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta 10Y \text{ Spread}$							0.127*** (0.013)	0.127*** (0.013)
$\Delta 3M \text{ Spread}$								0.008 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.002	0.019	0.038	0.046	0.048	0.048	0.087	0.086

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Taken together, the evidence from both periods shows that movements in $\Delta \log(\text{XRP}_{JPY})$ do not provide systematic predictive content for Japanese equity returns after conditioning on global and domestic macro financial factors. The patterns are consistent with a setting in which the Japanese equity risk premium responds primarily to global dollar conditions, risk sentiment, and the shape of the domestic yield curve, while Ripple remains on the periphery of this pricing environment. The wedge between the crypto leg and the local pricing kernel does not collapse in either period, and any superficial comovement that appears in the raw regressions is fully absorbed once the relevant global controls are included. As a result, Ripple does not evolve into a meaningful conduit for global risk transmission in Japan, even in the post 2020 era when other major crypto assets exhibit stronger integration with equity markets elsewhere.

To summarize, the Japanese results show that the post-2020 period is characterized by a consistent increase in the sensitivity of Japanese equities to the major non-pegged USD crypto assets, with Bitcoin and Ethereum displaying the largest transitions. Litecoin and Ripple exhibit more modest shifts, and their explanatory power remains limited relative to BTC and ETH, yet the direction of change is consistent across all assets. The evidence is therefore aligned with the model's prediction that as crypto markets mature and integrate into global dollar funding and risk-taking cycles, the crypto leg becomes more tightly linked to the pricing kernel used by global investors. In the Japanese context, where domestic financial conditions have historically been shaped by global risk appetite and exchange rate movements, this increased linkage is particularly salient. As crypto returns move more closely with the same dollar-based shocks that influence Japanese equities, the post-2020 period reveals a clearer and more systematic crypto-equity connection than what prevailed before the deepening of institutional participation and market liquidity.

4.8 China

In this subsection, we study whether USD-denominated crypto assets, once converted into renminbi, move systematically with Chinese equity markets.

Tables 41 and 42 present daily regressions of Chinese equity returns on CNY adjusted Bitcoin returns for the periods before and after 2020, with an expanding set of controls that account for movements in global commodities, the broad dollar index, implied volatility, and benchmark interest rates.

In the period before 2020, the estimated slope on $\Delta \log(\text{BTC}_{\text{CNY}})$ in Table 41 is small in magnitude (0.004 in column (1)) and declines further as controls are added, stabilizing at an economically minor value of approximately 0.003 in columns (4) to (8). The adjusted R^2 rises from essentially zero to about 0.031. This pattern suggests that prior to 2020 Bitcoin offered very limited incremental information for Chinese equity returns once common global drivers were accounted for. From the viewpoint of the model, a relatively large parity wedge in this period prevents the crypto leg from loading tightly on the stochastic discount factor that prices Chinese equities. Most short term variation in equity returns is instead absorbed by well known macro financial forces, particularly those linked to global risk sentiment and dollar funding conditions, leaving only a faint and statistically weak residual role for the crypto leg.

Table 41: Pre-2020 China Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{\text{CNY}})$	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.002** (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					-0.000 (0.005)	-0.000 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta 1\text{M Yield}$						-0.000 (0.010)	-0.000 (0.010)	-0.000 (0.010)
$\Delta 10\text{Y Spread}$							0.028*** (0.009)	0.033*** (0.009)
$\Delta 1\text{Y Spread}$								-0.012** (0.006)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.023	0.024	0.027	0.027	0.027	0.030	0.031

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 estimates in Table 42 indicate a partial change in this relationship. The bivariate regression shows a markedly larger and strongly significant slope of 0.020 (column (1)), suggesting that raw Bitcoin movements became more aligned with Chinese equity returns after 2020. Once the full block of controls is introduced, however, the coefficient attenuates to a small value of about 0.005 in columns (7) to (8). The adjusted R^2 increases from 0.005 to roughly 0.094 across the sequence of specifications. These results indicate that while Bitcoin began to co move more visibly with Chinese equities in the later period, the majority of this comovement is mediated through global risk factors. After accounting for commodity prices, the dollar, volatility, and rate movements, the residual predictive content of $\Delta \log(\text{BTC}_{\text{CNY}})$ is modest but remains positive, consistent with a slight compression of the parity wedge and a limited integration of crypto returns into the broader pricing environment facing Chinese investors.

The behavior of the square root variance ratio reinforces this interpretation. Both subsamples yield values close to unity, with $\sqrt{\rho} \approx 1.016$ in the period before 2020 and $\sqrt{\rho} \approx 1.035$ in the period after 2020. These values indicate that the residual wedge remains small relative to the idiosyncratic component of the crypto leg, although the slight increase after 2020 suggests a mild widening of the residual gap. Taken together, the results for China point to a gradual but incomplete transmission of crypto related shocks into domestic equity returns. While the post 2020 period sees a movement away from a pure hedge or low comovement regime toward a setting in which Bitcoin exhibits a modest positive association with Chinese equities, the explanatory weight of global financial conditions dominates in both windows.

Table 42: Post-2020 China Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{CNY})$	0.020*** (0.006)	0.014** (0.006)	0.008 (0.006)	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)	0.005 (0.006)	0.005 (0.006)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.094*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.005	0.026	0.055	0.055	0.056	0.056	0.089	0.094

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The Ethereum results in Tables 43 and 44 display a broadly similar structure to the Bitcoin findings, though with somewhat weaker magnitudes. In the pre-2020 window, the bivariate regression in column (1) yields a small and statistically insignificant loading on $\Delta \log(\text{ETH}_{CNY})$ of 0.003. As controls for commodities, the dollar, implied volatility, interest rates, and term structure variables are progressively introduced, the coefficient compresses toward zero and settles around an imprecise value of approximately 0.001 in columns (4) to (8). Over the same sequence of specifications, the adjusted R^2 rises from a slightly negative value to roughly 0.083. This pattern is consistent with a situation in which broad macrofinancial forces dominate the joint variation of the China equity index and Ethereum returns during this period. Once these global shocks are accounted for, little incremental explanatory power is left in the Ethereum leg. In the language of the model, the parity wedge remains sizable, and the crypto priced in renminbi does not load meaningfully on the pricing kernel that governs Chinese equity valuations.

Table 43: Pre-2020 China Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{\text{CNY}})$	0.003 (0.004)	0.002 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)
$\Delta \text{Commodity}$		0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
ΔDollar			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
ΔVIX				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.008)	0.022*** (0.008)	0.017** (0.008)	0.017** (0.008)
$\Delta 1\text{M Yield}$						-0.010 (0.010)	-0.009 (0.010)	-0.009 (0.010)
$\Delta 10\text{Y Spread}$							0.074*** (0.013)	0.074*** (0.013)
$\Delta 1\text{Y Spread}$								-0.001 (0.010)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	-0.001	0.047	0.052	0.052	0.057	0.057	0.084	0.083

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 44: Post-2020 China Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{\text{CNY}})$	0.022*** (0.005)	0.017*** (0.005)	0.012*** (0.005)	0.012** (0.005)	0.012** (0.005)	0.012** (0.005)	0.011** (0.005)	0.011** (0.005)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.094*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.010	0.030	0.057	0.057	0.059	0.058	0.091	0.097

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-2020 estimates tell a different story. The bivariate coefficient in column (1) is considerably larger at 0.022 and is strongly significant, indicating that raw comovement between Ethereum (in CNY terms) and Chinese equities becomes more visible once global investor appetite for crypto deepens and institutional participation broadens. This effect attenuates with the addition of controls but remains economically meaningful and statistically robust, settling around 0.011 in columns (7) and (8). The adjusted R^2 simultaneously increases from about 0.010 in the simple specification to nearly 0.097 in the fully controlled model. These results indicate that although a large share of the observed comovement continues to reflect common global

shocks, Ethereum retains a modest but persistent positive loading on Chinese equity returns even after controls are introduced. This pattern is consistent with the model's prediction that once the wedge shrinks due to deeper market integration, the crypto leg begins to move with the same pricing kernel that influences domestic equities.

The square root variance ratio further reinforces this interpretation. Both periods exhibit values near unity, with $\sqrt{\rho} \approx 1.008$ before 2020 and $\sqrt{\rho} \approx 1.012$ after 2020, indicating that the residual gap is small relative to the idiosyncratic crypto component in both windows. The slight increase in the post-2020 value suggests that although integration has improved, the Chinese market continues to display a somewhat larger residual dispersion compared with the advanced economies analyzed earlier. Nevertheless, the direction of change in the slope and the persistence of its significance even after extensive controls point toward a gradual transition from a hedge type or low comovement regime before 2020 to a modest but stable positive relation between Ethereum and Chinese equities thereafter. These findings complement the Bitcoin evidence and together suggest that in China, where financial conditions and capital flow management policies differ markedly from those in advanced economies, the integration of crypto into the domestic risk environment has increased but remains incomplete.

Continuing the analysis for China, the pre-2020 regressions for Litecoin show that as common shock controls are added, the slope remains small but persistently positive, moving only slightly from 0.014 in column (1) to approximately 0.013 in columns (7) to (8). Although the magnitude is limited, the loading survives a wide set of macro controls, and the adjusted R^2 increases from essentially zero to about 0.129. This pattern indicates that before 2020, Litecoin displayed a mild form of equity like comovement in renminbi terms, even once broad macro forces such as commodities, the dollar, volatility, interest rates, and term structure indicators were accounted for. The modest but durable positive coefficient suggests that a thin but nontrivial link existed between Chinese equities and the Litecoin leg, although the strength of this association remained overshadowed by the dominant role of common global shocks.

Table 45: Pre-2020 China Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{CNY})$	0.014* (0.007)	0.014** (0.007)	0.014** (0.007)	0.014** (0.007)	0.014** (0.007)	0.015** (0.007)	0.013* (0.007)	0.013* (0.007)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
ΔDollar			-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.003** (0.002)
ΔVIX				-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.027** (0.012)	0.028** (0.012)	0.013 (0.012)	0.013 (0.012)
$\Delta 1M \text{ Yield}$						-0.019 (0.015)	-0.018 (0.015)	-0.018 (0.015)
$\Delta 10Y \text{ Spread}$							0.126*** (0.019)	0.126*** (0.020)
$\Delta 1Y \text{ Spread}$								-0.000 (0.013)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.004	0.066	0.072	0.071	0.077	0.078	0.131	0.129

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 46: Post-2020 China Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{CNY})$	0.016*** (0.004)	0.013*** (0.004)	0.010** (0.004)	0.009** (0.004)	0.009** (0.004)	0.009** (0.004)	0.008** (0.004)	0.008** (0.004)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006* (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.093*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.007	0.028	0.057	0.056	0.058	0.058	0.090	0.096

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 period, the Litecoin coefficient in the bivariate specification is slightly larger at 0.016 in column (1), reflecting greater raw comovement during a time of more synchronized global financial conditions. Once controls are added, however, the coefficient contracts to approximately 0.009 in columns (4) to (6) and settles near 0.008 in the full specification (columns (7) to (8)). The adjusted R^2 improves from roughly 0.007 to about 0.096, indicating that the explanatory power of the full model rises modestly but remains driven primarily by the global factor block rather than by Litecoin itself. The results therefore suggest that much of the contemporaneous association between Litecoin and Chinese equities in the later period is mediated by broad co-movement with global macro conditions, leaving only a small residual role for the renminbi adjusted crypto leg.

Using the square root variance ratio, the evidence points to broadly similar residual parity in both windows, with $\sqrt{\rho} \approx 1.003$ before 2020 and ≈ 1.009 after 2020. Values near one imply that the unexplained wedge remains small relative to the idiosyncratic component of the crypto leg, and that the structural relationship between Litecoin and Chinese equities is not driven by changes in residual dispersion. The main difference across the two periods instead arises from changes in the overall fit of the model and in the sensitivity of Litecoin to global conditions. In both subperiods, the Chinese equity market absorbs most of the relevant information through common global shocks, while Litecoin contributes a small but consistently positive loading that becomes slightly less pronounced once the full set of controls is introduced.

A similar pattern is visible for Tether, although with dynamics that reflect its peg to the U.S. dollar rather than equity-related risk. In the pre-2020 regressions, the Tether slope starts at a modest negative value in the bivariate specification (-0.070 in col. 1) and moves toward a small, imprecise loading once common shocks are included (approximately -0.015 in cols. 7–8), while the adjusted R^2 increases from essentially zero to about 0.122. This behavior is consistent with a peg-neutral environment in which nearly all observable variation in $\Delta \log(\text{USDT}_{CNY})$ simply reflects dollar movements or short-lived deviations from the peg. Once broad macro forces such as commodities, dollar developments, volatility, and the yield curve are accounted for, Tether provides almost no incremental information about Chinese equity returns. In the language of the model, the crypto FX equity wedge is large in this period, and the Tether leg does not load meaningfully on the Chinese pricing kernel.

Table 47: Pre-2020 China Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{CNY})$	-0.070** (0.034)	-0.044 (0.034)	-0.034 (0.034)	-0.034 (0.034)	-0.031 (0.034)	-0.030 (0.034)	-0.015 (0.033)	-0.015 (0.033)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
ΔDollar			-0.004** (0.001)	-0.004** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.026*** (0.010)	0.027*** (0.010)	0.016* (0.010)	0.016* (0.010)
$\Delta 1\text{M Yield}$						-0.011 (0.012)	-0.012 (0.012)	-0.012 (0.012)
$\Delta 10\text{Y Spread}$							0.120*** (0.016)	0.121*** (0.016)
$\Delta 1\text{Y Spread}$								-0.004 (0.011)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.004	0.058	0.064	0.063	0.069	0.069	0.123	0.122

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 48: Post-2020 China Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{CNY})$	-0.741*** (0.075)	-0.647*** (0.077)	-0.447*** (0.086)	-0.449*** (0.086)	-0.455*** (0.086)	-0.456*** (0.086)	-0.389*** (0.085)	-0.385*** (0.085)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.009** (0.004)	0.009** (0.004)	0.007* (0.004)	0.006* (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.003)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.089*** (0.011)	0.100*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.044	0.055	0.066	0.066	0.069	0.068	0.097	0.103

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 sample, however, the bivariate relation becomes sharply negative, with a sizeable coefficient of -0.741 in col. 1. This magnitude decreases once controls enter the specification, but the loading remains economically meaningful at approximately -0.389 in cols. 7–8. Despite this strong unconditional relation, the adjusted R^2 rises only modestly from 0.044 to about 0.103 when controls are added, which suggests that the observed comovement is dominated by macro financial forces rather than by a genuine Tether equity linkage. The square root variance ratio reinforces this interpretation: residual parity becomes much looser across windows, with $\sqrt{\rho} \approx 1.401$ before 2020 and a markedly larger $\sqrt{\rho} \approx 4.116$ after 2020. Such a widening residual gap is consistent with peg mechanics and conversion noise rather than a structural

link to equity pricing. Consequently, the fully controlled estimates offer the most reliable signal, indicating that once global conditions are absorbed, Tether remains largely unrelated to Chinese equity dynamics. The post-2020 period therefore represents not a strengthening of crypto equity integration but an amplification of transitory peg deviations that enter the data as noise rather than as priced risk.

Finally, a closely related pattern appears for Ripple in the Chinese market. The pre 2020 regressions indicate a near neutral relation between Ripple returns and domestic equity performance. The bivariate slope is small (0.014 in col. 1) and remains of similar magnitude once the full set of macro financial controls is included (approximately 0.015 in cols. 7–8). At the same time, the adjusted R^2 improves from essentially zero to about 0.130 as movements in commodities, the dollar, volatility, and the interest rate structure absorb most of the variation in equity returns. This behaviour is consistent with a setting in which the parity wedge between the Ripple leg and the equity pricing kernel remains large, so any raw comovement is largely accounted for by common global shocks rather than by a systematic linkage between Ripple and Chinese equities.

Table 49: Pre-2020 China Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{CNY})$	0.014* (0.008)	0.015* (0.008)	0.015* (0.008)	0.014* (0.008)	0.014* (0.008)	0.015* (0.008)	0.015** (0.007)	0.015** (0.007)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
ΔDollar			-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.003** (0.002)
ΔVIX				-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.027** (0.012)	0.028** (0.012)	0.013 (0.012)	0.013 (0.012)
$\Delta 1\text{M Yield}$						-0.019 (0.015)	-0.018 (0.015)	-0.018 (0.015)
$\Delta 10\text{Y Spread}$							0.127*** (0.019)	0.127*** (0.020)
$\Delta 1\text{Y Spread}$								0.000 (0.013)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.003	0.065	0.072	0.071	0.076	0.077	0.131	0.130

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 evidence follows a similar structure. The bivariate regression shows a modest but significant loading of 0.012 (col. 1), yet this effect diminishes after the full set of controls is added, declining to approximately 0.008 in cols. 4–8. Model fit rises from 0.005 to roughly 0.096, confirming that the explanatory power in the later period is again primarily driven by standard global risk variables rather than by Ripple specific movements.

The square root variance ratio $\sqrt{\rho}$ reinforces this interpretation, remaining close to unity across both periods ($\sqrt{\rho} \approx 1.004$ pre 2020 and ≈ 1.005 post 2020). These values signal that the residual wedge is small relative to the idiosyncratic crypto component, yet the controlled beta remains economically negligible. Overall, the Chinese results reveal a stable low comovement structure in which changes in $\Delta \log(\text{XRP}_{CNY})$ provide little incremental information for domestic equity returns once global macro conditions are taken into account.

Table 50: Post-2020 China Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{CNY})$	0.012*** (0.004)	0.010*** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006* (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.094*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.005	0.027	0.057	0.056	0.058	0.058	0.090	0.096

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Across assets, China offers the weakest evidence of a post-2020 crypto FX equity linkage among the markets studied. Ethereum is the only token that shows a clear strengthening in equity-like behavior, with its fully controlled slope rising from essentially zero before 2020 to about 0.011 afterward. Bitcoin displays only a modest increase (from roughly 0.003 to 0.005), while the corresponding loadings for Litecoin and Ripple actually decline once controls are included (from about 0.013 to 0.008 for LTC and from about 0.015 to 0.008 for XRP). Tether exhibits a pronounced negative coefficient in the controlled specification (around -0.39), consistent with its peg structure and the presence of policy frictions rather than any systematic equity exposure. The gains in adjusted R^2 remain limited throughout, reinforcing that the Chinese case reflects a fragmented and asset-specific pattern rather than a unified shift toward stronger equity like comovement. These results fit the broader emerging market narrative in which crypto and local equities are linked only weakly and in uneven ways, with much of the interaction shaped by exchange rate movements, peg mechanisms, and country-level frictions instead of a stable and shared pricing kernel.

4.9 Brazil

In this subsection we examine whether USD-denominated crypto assets, once converted into Brazilian reals, move systematically with Brazilian equity markets.

For Bitcoin, the pre-2020 regressions in Table 51 show limited evidence of systematic comovement. The bivariate slope on $\Delta \log(\text{BTC}_{BRL})$ is slightly negative (-0.005 in col. 1) and converges toward zero once controls for commodities, the dollar, volatility, interest rates, and term spreads are introduced (cols. 7–8). Over the same sequence, the adjusted R^2 increases from essentially zero to about 0.322. This pattern suggests that before 2020, the return dynamics of Brazilian equities were almost entirely driven by broad global and domestic macro forces, leaving the FX adjusted Bitcoin leg with little marginal explanatory power. Such behavior is consistent with the model’s prediction that when the crypto to FX to equity wedge remains large, the crypto leg does not load tightly on the pricing kernel that governs domestic equity returns.

Table 51: Pre-2020 Brazil Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{BRL})$	-0.005 (0.004)	-0.004 (0.003)	-0.002 (0.003)	-0.004 (0.003)	-0.004 (0.003)	-0.004 (0.003)	0.001 (0.003)	0.001 (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.012** (0.005)	0.012** (0.005)	0.021*** (0.004)	0.022*** (0.004)
$\Delta 1\text{M Yield}$						0.002 (0.009)	-0.001 (0.009)	-0.004 (0.009)
$\Delta 10\text{Y Spread}$							-0.029*** (0.002)	-0.025*** (0.002)
$\Delta 1\text{Y Spread}$								-0.015*** (0.003)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.102	0.138	0.239	0.241	0.240	0.316	0.322

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 52: Post-2020 Brazil Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{BRL})$	0.048*** (0.009)	0.040*** (0.009)	0.032*** (0.009)	-0.015* (0.008)	-0.016** (0.008)	-0.016** (0.008)	0.000 (0.007)	-0.000 (0.007)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.021*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1\text{Y Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.013	0.064	0.113	0.301	0.307	0.308	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 sample, the raw correlation between Bitcoin and Brazilian equities appears stronger. The bivariate coefficient is sizable and positive (0.048 in col. 1), indicating that periods of strong crypto performance were associated with higher domestic equity returns. However, once common shocks are absorbed, the Bitcoin slope attenuates sharply, turning modestly negative in the middle specifications (about -0.016 in cols. 4–6) and ending close to zero once the yield curve variables are introduced (cols. 7–8). Model fit rises markedly across specifications (Adj. R^2 : 0.013 to ~ 0.444). These results imply that most of the apparent strengthening in the headline Bitcoin–equity correlation after 2020 is mediated by broad global

factors rather than by a robust independent Bitcoin loading.

The square root variance ratio $\sqrt{\rho}$ reinforces this interpretation. The pre-2020 window lies close to one ($\sqrt{\rho} \approx 1.014$), while the post-2020 value rises moderately above one (≈ 1.051), indicating that the residual wedge remains meaningful and even becomes somewhat larger in the later sample. This suggests that the parity condition between the crypto leg and Brazilian equities remains relatively loose, consistent with a setting in which exchange rate movements, macro financial volatility, and country-specific frictions do most of the explanatory work, while the independent role of Bitcoin in shaping Brazilian equity returns remains limited.

Ethereum exhibits a broadly similar structure. The pre-2020 regressions indicate that once common shock controls are introduced, the Ethereum slope becomes more negative, moving from a modest bivariate estimate of -0.009 in column (1) to roughly -0.014 in columns (4) to (6). When term spread variables enter, the coefficient attenuates toward zero, settling between -0.005 and -0.006 in columns (7) and (8). Over the same sequence, model fit increases notably, with the adjusted R^2 rising from 0.002 to approximately 0.416. This pattern is consistent with a pre-2020 environment in which fluctuations in commodities, the dollar, implied volatility, interest rates, and the structure of yield spreads account for most movements in Brazilian equities. Once these broad macro forces are controlled for, the Ethereum return in Brazilian reais contributes little independent information, aside from a small negative association that weakens once the term structure is taken into account.

Table 53: Pre-2020 Ethereum Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{BRL})$	-0.009 (0.005)	-0.009* (0.005)	-0.011** (0.005)	-0.014*** (0.005)	-0.014*** (0.005)	-0.014*** (0.005)	-0.005 (0.004)	-0.006 (0.004)
$\Delta \text{Commodity}$		0.006*** (0.001)	0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
ΔDollar			-0.007*** (0.001)	-0.006*** (0.001)	-0.007*** (0.001)	-0.006*** (0.001)	-0.003** (0.001)	-0.002** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016* (0.009)	0.016* (0.009)	0.024*** (0.008)	0.023*** (0.008)
$\Delta 1\text{M Yield}$						-0.007 (0.012)	-0.013 (0.010)	-0.013 (0.010)
$\Delta 10\text{Y Spread}$							-0.048*** (0.002)	-0.035*** (0.003)
$\Delta 1\text{Y Spread}$								-0.026*** (0.004)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.002	0.077	0.102	0.173	0.174	0.174	0.396	0.416

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

After 2020, the bivariate regression displays a sizable and statistically strong positive loading on $\Delta \log(\text{ETH}_{BRL})$ (0.047 in column 1). However, once the standard global and domestic controls are included, the coefficient contracts substantially. With the core control set (columns 4 to 6), the estimate becomes small and statistically indistinguishable from zero (approximately -0.008), and once term spreads enter in columns 7 and (8), the slope settles near zero. Despite this attenuation, the adjusted R^2 improves from 0.022 in the bivariate specification to roughly 0.444 with full controls, indicating that most of the explanatory gain comes from macro financial variables rather than from Ethereum itself.

Table 54: Post-2020 Ethereum Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{BRL})$	0.047*** (0.007)	0.040*** (0.007)	0.032*** (0.007)	-0.008 (0.006)	-0.008 (0.006)	-0.008 (0.006)	0.002 (0.005)	0.001 (0.005)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.021*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1\text{Y Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.022	0.070	0.117	0.301	0.307	0.307	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The square root variance ratios support this interpretation. Both windows lie near, but above, unity, with $\sqrt{\rho} \approx 1.015$ before 2020 and $\sqrt{\rho} \approx 1.029$ after 2020. These values imply that the residual wedge remains sizable relative to the idiosyncratic Ethereum component, and that the parity relation between the FX adjusted crypto leg and equity returns does not tighten in the later period. The mild increase in $\sqrt{\rho}$ after 2020 suggests a slightly wider residual gap, consistent with an environment in which Ethereum returns in Brazilian reais remain largely dominated by broader global forces rather than forming a meaningful standalone equity linked channel. Together, the evidence indicates that the raw post2020 ETH–equity correlation is mostly absorbed once common shocks are removed, leaving no stable or robust Ethereum loading in the Brazilian case.

A comparable pattern appears for Ripple in the Brazilian market. The pre-2020 regressions present a pattern of weak and statistically imprecise comovement between Litecoin returns and Brazilian equities. The bivariate coefficient on $\Delta \log(\text{LTC}_{BRL})$ is essentially zero (-0.003 in col. 1) and remains economically negligible once controls for common shocks are introduced. As global drivers such as commodities, the broad dollar, volatility, interest rates, and the domestic term structure are added, the slope drifts toward values very close to zero (cols. 4 to 6) and settles at about 0.005 to 0.006 when term spreads are included (cols. 7 to 8). Over the same sequence, overall fit improves markedly (Adj. R^2 : $-0.001 \rightarrow 0.382$). These results are consistent with a pre-2020 environment in which most variation in Brazilian equity returns reflects broad macro financial forces rather than the behavior of Litecoin. In the language of the model, the crypto–FX–equity wedge is large in this period and prevents Litecoin from loading on the local pricing kernel, leaving little systematic linkage beyond exposure to shared global factors.

Table 55: Pre-2020 Brazil Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{BRL})$	-0.003 (0.008)	-0.001 (0.008)	-0.001 (0.007)	-0.002 (0.007)	-0.002 (0.007)	-0.002 (0.007)	0.005 (0.006)	0.006 (0.006)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.003** (0.002)	-0.003* (0.002)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10Y \text{ Yield}$					0.011 (0.012)	0.013 (0.012)	0.021* (0.011)	0.018* (0.010)
$\Delta 1M \text{ Yield}$						-0.017 (0.015)	-0.017 (0.013)	-0.017 (0.013)
$\Delta 10Y \text{ Spread}$							-0.048*** (0.003)	-0.036*** (0.004)
$\Delta 1Y \text{ Spread}$								-0.028*** (0.005)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.061	0.087	0.160	0.160	0.161	0.355	0.382

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 56: Post-2020 Brazil Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{BRL})$	0.031*** (0.006)	0.026*** (0.006)	0.022*** (0.006)	-0.008 (0.005)	-0.008 (0.005)	-0.008 (0.005)	0.001 (0.005)	0.000 (0.005)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10Y \text{ Yield}$					0.020*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1M \text{ Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10Y \text{ Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1Y \text{ Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.012	0.063	0.113	0.301	0.307	0.307	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 window, the unconditional association strengthens at first glance: the bivariate coefficient rises to 0.031 (col. 1), suggesting a positive raw link between Brazilian equities and Litecoin once converted into reals. However, the effect dissipates almost entirely once common shocks are absorbed. With core controls, the slope becomes negative (about -0.008 in cols. 4 to 6), and once the term structure is added, the coefficient collapses to values effectively equal to zero (cols. 7 to 8). Over the same specifications, the adjusted R^2 rises from 0.012 to approximately 0.444, indicating that improvements in fit are driven by broad global and domestic financial conditions rather than by any robust Litecoin contribution. This disconnect

between the larger bivariate coefficient and the negligible fully controlled effect indicates that Litecoin's apparent post-2020 comovement with Brazilian equities is almost entirely mediated by shared macro forces rather than a standalone crypto–equity exposure.

The square root variance ratio reinforces this interpretation. Both periods lie close to unity ($\sqrt{\rho} \approx 1.007$ pre 2020; ≈ 1.023 post 2020), indicating that the residual wedge remains of similar magnitude relative to the idiosyncratic crypto component, with a slightly looser parity in the later window. Combined with the weak and non robust slope estimates, this evidence points to a muted and largely indirect role for $\Delta \log(\text{LTC}_{BRL})$ in explaining Brazilian equity returns, even after the broader global transition toward stronger crypto–equity comovement in advanced markets after 2020.

The Tether results for Brazil reveal a clear and economically intuitive pattern that differs sharply from the behavior of nonpegged crypto assets. In both periods, the estimated slope on $\Delta \log(\text{USDT}_{BRL})$ is negative and sizable, indicating that increases in the USD value of Tether relative to the real are associated with lower equity returns. This is consistent with a currency risk channel rather than a crypto-specific risk premium. When the real depreciates, the domestic cost of dollar-linked assets rises, and equity valuations tend to fall, producing a persistent inverse relation between USDT returns and Brazilian equities.

In the pre-2020 window, the bivariate regression produces a large negative coefficient (-0.264 in column 1), and although the magnitude declines once controls for commodities, the dollar, volatility, interest rates, and term structure are introduced, the slope remains negative and economically meaningful at roughly -0.05 in the fully controlled specification (columns 7 and 8). Model fit improves substantially over the sequence of specifications, with the adjusted R^2 rising from 0.085 to approximately 0.386. These patterns indicate that before 2020, the USDT return primarily captured movements in the bilateral exchange rate and that these currency fluctuations played a nontrivial role in explaining daily equity variation even after absorbing broad global risk factors.

Table 57: Pre-2020 Brazil Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{BRL})$	-0.264*** (0.029)	-0.227*** (0.029)	-0.211*** (0.029)	-0.195*** (0.028)	-0.197*** (0.028)	-0.196*** (0.028)	-0.057** (0.027)	-0.049* (0.026)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.003** (0.001)	-0.003** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.021** (0.010)	0.022** (0.010)	0.028*** (0.009)	0.026*** (0.009)
$\Delta 1\text{M Yield}$						-0.008 (0.013)	-0.010 (0.011)	-0.010 (0.011)
$\Delta 10\text{Y Spread}$							-0.046*** (0.003)	-0.036*** (0.004)
$\Delta 1\text{Y Spread}$								-0.027*** (0.005)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.085	0.129	0.145	0.220	0.223	0.223	0.366	0.386

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-2020 period amplifies this relation. The bivariate coefficient becomes very large and negative (-0.783 in column 1), and although controls attenuate some of the raw co movement, the fully controlled estimate remains strongly negative at approximately -0.37 (columns 7 and 8). Model fit also rises meaningfully, with the adjusted R^2 increasing from 0.253 to around 0.482. These results suggest that after 2020, exchange rate volatility and dollar funding conditions became even more tightly linked to equity movements in Brazil, and that USDT returns effectively summarize this stronger currency based transmission mechanism.

Table 58: Post-2020 Brazil Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{BRL})$	-0.783*** (0.029)	-0.738*** (0.030)	-0.688*** (0.032)	-0.557*** (0.029)	-0.560*** (0.029)	-0.560*** (0.029)	-0.377*** (0.030)	-0.367*** (0.030)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001* (0.000)	0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.004*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.004)	0.021*** (0.004)	0.031*** (0.004)	0.031*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.004 (0.004)
$\Delta 10\text{Y Spread}$							-0.030*** (0.002)	-0.026*** (0.002)
$\Delta 1\text{Y Spread}$								-0.014*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.253	0.262	0.271	0.403	0.410	0.410	0.475	0.482

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The square root variance ratio supports this interpretation. The pre-2020 estimate lies above unity at $\sqrt{\rho} \approx 1.308$, while the post-2020 value increases further to approximately 1.924, indicating that the residual wedge relative to the pure crypto leg becomes wider over time. This wider gap is consistent with the role of Tether as a pegged instrument whose variation reflects currency pressure rather than a crypto-specific shock. Together, these estimates underscore that in Brazil, Tether behaves primarily as an exchange rate factor rather than an equity-like exposure, with the currency-equity linkage becoming more pronounced after 2020.

A similar pattern emerges when examining Ripple. The pre-2020 regressions show weak and imprecise comovement between Ripple and Brazilian equities. The bivariate specification produces a coefficient that is effectively zero (-0.000 in col. 1), and the slope remains economically negligible once common shock controls are added, fluctuating around zero and reaching only about 0.003 once term spreads enter the specification (cols. 7–8). Model fit nevertheless increases substantially as controls are incorporated, with the adjusted R^2 rising from -0.001 to 0.381. These results indicate that, before 2020, broad macro forces such as movements in commodities, the dollar, volatility, interest rates, and the term structure explain nearly all equity variation in Brazil, leaving very little incremental role for $\Delta \log(\text{XRP}_{BRL})$. This pattern is consistent with a low comovement environment in which the FX-converted Ripple leg does not load meaningfully on the pricing kernel relevant for Brazilian equities.

Table 59: Pre-2020 Brazil Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{BRL})$	-0.000 (0.008)	0.002 (0.008)	0.002 (0.008)	0.000 (0.008)	0.000 (0.008)	0.001 (0.008)	0.003 (0.007)	0.003 (0.007)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.003** (0.002)	-0.003* (0.002)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10Y \text{ Yield}$					0.011 (0.012)	0.013 (0.012)	0.021* (0.011)	0.018* (0.010)
$\Delta 1M \text{ Yield}$						-0.017 (0.015)	-0.017 (0.013)	-0.017 (0.013)
$\Delta 10Y \text{ Spread}$							-0.048*** (0.003)	-0.036*** (0.004)
$\Delta 1Y \text{ Spread}$								-0.028*** (0.005)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.061	0.087	0.160	0.160	0.161	0.355	0.381

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 60: Post-2020 Brazil Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{BRL})$	0.016*** (0.005)	0.013** (0.005)	0.012** (0.005)	-0.005 (0.005)	-0.005 (0.005)	-0.005 (0.005)	0.002 (0.004)	0.002 (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10Y \text{ Yield}$					0.020*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1M \text{ Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10Y \text{ Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1Y \text{ Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.004	0.057	0.110	0.300	0.306	0.307	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 sample, the bivariate Ripple coefficient rises modestly to 0.016 (col. 1), suggesting some raw comovement between XRP and Brazilian equities. However, this relationship weakens sharply once controls are introduced: the coefficient becomes slightly negative with core controls (around -0.005 in cols. 4–6) and then returns to a value near zero once term spreads are included (approximately 0.002 in cols. 7–8). At the same time, the adjusted R^2 improves from 0.004 to about 0.444, indicating that the explanatory content of the regressions comes almost entirely from common macro financial drivers rather than from XRP itself. These dynamics point to a setting in which any apparent post-2020 comovement is

largely absorbed by global factors, and the FX adjusted Ripple return offers almost no independent predictive power for Brazilian equity movements after conditioning on broader risk conditions.

The square-root variance ratio $\sqrt{\rho}$ remains close to one in both periods, with values of approximately 1.012 before 2020 and 1.015 after 2020. These figures imply a slightly wider residual gap in the later window, but the overall parity remains relatively tight when compared to the idiosyncratic component of the crypto leg. The stability of $\sqrt{\rho}$ reinforces that the principal development over time is not a shift in residual dispersion but rather that XRP continues to play almost no systematic role in explaining equity returns once common shocks are taken into account.

To summarize, the fully controlled specifications in column 8 show even less evidence of a strong or stable post 2020 crypto equity loading in Brazil than in China. The BTC coefficient moves from a small positive value of 0.001 before 2020 to essentially zero afterwards, while the ETH coefficient changes from a mildly negative value -0.006 to a statistically negligible 0.001. The corresponding LTC loading declines from 0.006 to 0.000, and the XRP coefficient from 0.003 to 0.002, indicating that none of the major non pegged assets acquire a systematic role once the full set of macro financial controls is included. In contrast, USDT becomes sharply more negative, falling from approximately -0.049 before 2020 to roughly -0.367 afterwards, a pattern that reflects the interaction between the stablecoin peg, dollar funding pressure, and domestic market frictions rather than any form of pro-equity exposure. Taken together, the column 8 evidence fits our emerging-market narrative: Brazil’s post-2020 crypto–FX relationship remains weak, asset-specific, and dominated by exchange-rate and country risk—especially via USDT—rather than a single tight, equity-like crypto loading.

4.10 India

In this subsection, we study whether USD-denominated crypto assets, once converted into Indian rupees, move systematically with Indian equity markets.

The pre-2020 regressions for Bitcoin indicate a stable negative association with equities: the bivariate slope is modest but statistically precise at -0.008 (col. 1), and becomes slightly more negative once common shock controls are added, settling around -0.009 in the fully controlled specifications (cols. 4–8). The adjusted R^2 increases from essentially zero to about 0.088 as broad macro variables enter the regression. This pattern suggests that, before 2020, fluctuations in commodities, the dollar, volatility, and both short and long interest rate movements accounted for most of the variation in Indian equity returns, leaving Bitcoin with a small but persistent offsetting relation that is consistent with a low comovement or partial hedge profile. The fact that the slope remains negative even after controlling for global conditions is consistent with a setting in which the crypto–FX–equity wedge is large and Bitcoin does not load meaningfully on the pricing kernel that governs Indian stock returns.

Table 61: Pre-2020 India Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{\text{INR}})$	-0.008*** (0.002)	-0.008*** (0.002)	-0.008*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.002*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.006* (0.003)	0.006 (0.003)	0.006* (0.003)	0.006* (0.003)
$\Delta 1\text{M Yield}$						0.007 (0.007)	0.007 (0.007)	0.007 (0.007)
$\Delta 10\text{Y Spread}$							-0.025*** (0.003)	-0.022*** (0.003)
$\Delta 2\text{Y Spread}$								-0.005* (0.003)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.003	0.041	0.049	0.069	0.070	0.070	0.087	0.088

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 62: Post-2020 India Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{\text{INR}})$	0.028*** (0.007)	0.023*** (0.007)	0.010 (0.007)	-0.008 (0.007)	-0.008 (0.007)	-0.008 (0.007)	-0.008 (0.007)	-0.007 (0.007)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.002 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.019*** (0.007)	-0.012 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.008	0.018	0.101	0.132	0.140	0.140	0.142	0.144

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

After 2020, the Bitcoin coefficient becomes positive in the raw specification (0.028, col. 1), suggesting stronger contemporaneous comovement between INR-adjusted crypto returns and Indian equities. However, once the standard controls are introduced, this loading contracts sharply and becomes slightly negative and statistically weak, staying close to -0.008 to -0.007 in the fully controlled models (cols. 4–8). Meanwhile, model fit improves from 0.008 to approximately 0.144. This indicates that any apparent strengthening of the Bitcoin–equity relation in the raw data is largely driven by global risk forces rather than a structural crypto contribution. The square root variance ratio supports this interpretation: $\sqrt{\rho}$ is already close to one before 2020 (about 1.018) and rises moderately after 2020 (to about 1.053), revealing a slightly wider residual gap

in the later period and consistent with modest crypto-specific noise that prevents a strong equity-like loading from emerging.

Turning to Ethereum, the pre 2020 regressions show a similar pattern of weak and slightly negative parity. The bivariate coefficient is -0.007 (col. 1) and becomes marginally more negative, reaching roughly -0.009 , after the introduction of commodity, dollar, volatility, and interest rate controls (cols. 4–8). The adjusted R^2 rises from 0.003 to about 0.031. This points to a pre 2020 environment in which Indian equity returns were primarily shaped by global macro financial conditions, leaving little space for independent contribution from INR adjusted Ethereum returns, which exhibit a small offsetting sign consistent with limited integration into the domestic pricing kernel.

Table 63: Pre-2020 India Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{INR})$	-0.007** (0.003)	-0.008** (0.003)	-0.008** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001** (0.000)	0.001* (0.000)	0.001** (0.000)	0.001** (0.000)	0.001** (0.000)	0.001** (0.000)
ΔDollar			-0.003*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
ΔVIX				-0.000*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
$\Delta 10\text{Y Yield}$					-0.004 (0.006)	-0.004 (0.006)	-0.003 (0.006)	-0.003 (0.006)
$\Delta 1\text{M Yield}$						0.002 (0.008)	0.002 (0.008)	0.002 (0.008)
$\Delta 10\text{Y Spread}$							-0.004 (0.005)	-0.002 (0.005)
$\Delta 2\text{Y Spread}$								-0.006 (0.006)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.003	0.017	0.025	0.032	0.032	0.031	0.031	0.031

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post 2020 results show a noticeably larger bivariate loading for Ethereum (0.033, col. 1), suggesting an increase in raw comovement similar to what is observed for Bitcoin. Yet this loading is largely absorbed once controls are added, compressing to a small and statistically negligible value of about 0.006 (cols. 4–8). Adjusted R^2 again rises substantially, from 0.020 to around 0.144, indicating that global shocks and interest rate dynamics explain most of the comovement in the later period.

The square root variance ratio, which moves only slightly from $\sqrt{\rho} \approx 1.015$ before 2020 to $\sqrt{\rho} \approx 1.020$ after 2020, reveals similar residual parity in both subsamples. The modest uptick represents only a limited widening of the residual wedge and confirms that the principal shift is a stronger bivariate association that disappears once common global influences are controlled for. In economic terms, Ethereum exhibits a mild transition from a hedge like role toward a more positive but still modest equity linked pattern, though this linkage remains dominated by broader macro forces rather than by intrinsic crypto risk.

Table 64: Post-2020 India Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{INR})$	0.033*** (0.005)	0.030*** (0.005)	0.019*** (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.008*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.018*** (0.007)	-0.011 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.020	0.029	0.107	0.133	0.140	0.140	0.142	0.144

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The Ripple results fit the same overall pattern observed for Litecoin in the Indian market. The pre-2020 regressions indicate weak and slightly negative comovement between Litecoin and Indian equities. The bivariate slope is small and remains essentially unchanged at -0.006 across all specifications (cols. 1–8, SE ≈ 0.005), while the adjusted R^2 increases only modestly from 0.000 to 0.038 as controls for commodities, the dollar, implied volatility, interest rates, and term structure are added. This pattern suggests that before 2020 broad macroeconomic forces accounted for most movements in Indian equities, leaving $\Delta \log(\text{LTC}_{INR})$ with little explanatory power and a slightly offsetting sign. Such a configuration is consistent with a low comovement regime in which the crypto leg does not load meaningfully on the Indian pricing kernel and any apparent association is largely absorbed by standard global variables.

Table 65: Pre-2020 India Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{INR})$	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.001** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					-0.013 (0.009)	-0.014 (0.009)	-0.013 (0.009)	-0.012 (0.009)
$\Delta 1\text{M Yield}$						0.008 (0.011)	0.008 (0.011)	0.008 (0.011)
$\Delta 10\text{Y Spread}$							-0.006 (0.007)	-0.003 (0.007)
$\Delta 2\text{Y Spread}$								-0.009 (0.008)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.000	0.012	0.030	0.036	0.038	0.037	0.037	0.037

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 66: Post-2020 India Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{INR})$	0.014*** (0.005)	0.011** (0.005)	0.005 (0.004)	-0.006 (0.004)	-0.006 (0.004)	-0.006 (0.004)	-0.005 (0.004)	-0.005 (0.004)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.018*** (0.007)	-0.012 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.004	0.016	0.101	0.133	0.140	0.140	0.142	0.144

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 period, Litecoin shows a stronger unconditional association with Indian equities in the bivariate regression, where the coefficient rises to 0.014 (col. 1) and is statistically significant. Once controls are introduced, however, this loading dissipates and converges back to a small negative value of about -0.006 in the fully specified model (cols. 4–8). The adjusted R^2 nonetheless increases from 0.004 to approximately 0.144 as the control set captures common global shocks and domestic interest rate conditions. The disappearance of the Litecoin coefficient once controls are included indicates that most of the raw post-2020 association reflects the influence of macroeconomic and financial conditions rather than a structural

crypto–equity link in rupee terms.

The square root variance ratio provides an additional diagnostic for the strength of crypto–equity parity. Both windows yield values close to unity ($\sqrt{\rho} \approx 1.015$ pre2020; ≈ 1.025 post-2020), implying that the variance of the residual parity wedge remains small relative to the idiosyncratic crypto leg. The slight increase in the ratio in the later period points to a marginally wider residual gap, but not one large enough to generate a substantive Litecoin loading. Overall, the evidence aligns with the broader emerging market narrative: once common global shocks are removed, Litecoin contributes little to the variation in Indian equity returns, and post-2020 changes do not translate into a stable or economically meaningful crypto–equity linkage.

A broadly consistent pattern appears in the Tether estimates. The pre-2020 Tether regressions display a clear and persistent negative association with Indian equities. The bivariate specification yields a moderately sized inverse loading of -0.105 (col. 1), and this coefficient remains stable once controls for commodities, the dollar, volatility, interest rates, and term structure are added, settling around -0.093 to -0.094 in cols. 4 to 8. During this earlier period, the adjusted R^2 increases only modestly, from 0.020 to 0.047, indicating that most variation in equity returns is still captured by broad macro financial forces. Tether’s residual negative coefficient is therefore best interpreted as reflecting small, non-fundamental deviations of the peg that generate mild currency-related return differentials rather than a genuine systematic exposure to the Indian equity pricing kernel.

Table 67: Pre-2020 India Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{INR})$	-0.105*** (0.024)	-0.099*** (0.024)	-0.092*** (0.024)	-0.093*** (0.024)	-0.094*** (0.024)	-0.094*** (0.024)	-0.093*** (0.024)	-0.094*** (0.024)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001** (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)
ΔDollar			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003** (0.001)
ΔVIX				-0.000** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
$\Delta 10\text{Y Yield}$					-0.009 (0.007)	-0.010 (0.007)	-0.009 (0.007)	-0.009 (0.007)
$\Delta 1\text{M Yield}$						0.006 (0.009)	0.006 (0.009)	0.006 (0.009)
$\Delta 10\text{Y Spread}$							-0.003 (0.006)	-0.001 (0.006)
$\Delta 2\text{Y Spread}$								-0.008 (0.007)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.020	0.034	0.042	0.048	0.048	0.048	0.047	0.047

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 68: Post-2020 India Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{INR})$	-1.578*** (0.079)	-1.549*** (0.079)	-1.316*** (0.081)	-1.309*** (0.080)	-1.285*** (0.080)	-1.285*** (0.080)	-1.280*** (0.081)	-1.273*** (0.081)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.011*** (0.004)	0.011*** (0.004)	0.011*** (0.004)	0.011*** (0.004)
$\Delta 1\text{M Yield}$						0.002 (0.004)	0.002 (0.004)	0.002 (0.004)
$\Delta 10\text{Y Spread}$							-0.002 (0.007)	0.002 (0.007)
$\Delta 2\text{Y Spread}$								-0.008 (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.159	0.166	0.201	0.231	0.234	0.234	0.234	0.234

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

After 2020, the magnitude of this inverse relation becomes substantially larger. The bivariate slope reaches -1.578 in col. 1, and although attenuated by controls, it remains strongly negative with full conditioning, at approximately -1.28 in cols. 7 and 8. The adjusted R^2 rises from 0.159 to roughly 0.234 over this sequence, indicating that even after accounting for standard global and domestic factors, movements in the dollar-linked stablecoin continue to capture a sizeable inverse component of Indian equity variation. This pattern is consistent with increased sensitivity of the rupee to capital flow pressures, tighter policy constraints, and episodes in which deviations from the peg reflect heightened demand for dollar liquidity. In this environment, Tether returns in rupee terms exhibit a pronounced negative relation with domestic equity performance.

The square root variance ratio reinforces this interpretation. Its value sits away from unity in both windows, with $\sqrt{\rho} \approx 1.299$ before 2020 and an even larger ≈ 4.339 after 2020. These figures indicate that the residual parity wedge is sizable in both periods and widens considerably in the later sample, suggesting that non-fundamental forces, including currency market segmentation and liquidity driven flow dynamics, contribute meaningfully to the observed co-movement. In combination with the regression coefficients, the evidence points to a persistent and economically large inverse relationship between FX-adjusted USDT returns and Indian equities, with that relationship strengthening notably in the post-2020 period.

Finally, the pre-2020 regressions show weak and slightly negative comovement: the XRP slope begins at a small value in the bivariate specification (-0.008 , col. 1) and settles around -0.009 once common-shock controls are included (cols. 4–8), while overall fit improves modestly (Adj. R^2 : $0.002 \rightarrow 0.038$). This pattern indicates that before 2020 most variation in Indian equities was driven by broad macro forces such as commodities, the dollar, volatility, and interest rates, leaving $\Delta \log(\text{XRP}_{INR})$ with a limited and slightly offsetting relation to local stock returns. The negative sign is consistent with a hedge or diversifier type behavior in an environment where domestic and global risk factors dominate equity dynamics.

Table 69: Pre-2020 India Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{INR})$	-0.008 (0.006)	-0.008 (0.006)	-0.008 (0.005)	-0.009 (0.005)	-0.009 (0.005)	-0.009 (0.005)	-0.009 (0.005)	-0.009 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.001** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					-0.013 (0.009)	-0.014 (0.009)	-0.013 (0.009)	-0.012 (0.009)
$\Delta 1\text{M Yield}$						0.008 (0.011)	0.008 (0.011)	0.008 (0.011)
$\Delta 10\text{Y Spread}$							-0.006 (0.007)	-0.003 (0.007)
$\Delta 2\text{Y Spread}$								-0.009 (0.008)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.002	0.014	0.031	0.037	0.039	0.038	0.038	0.038

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 70: Post-2020 India Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{INR})$	0.004 (0.004)	0.002 (0.004)	-0.001 (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.008** (0.004)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.019*** (0.007)	-0.012 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	-0.000	0.013	0.100	0.134	0.141	0.141	0.143	0.145

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the period after 2020, the bivariate XRP coefficient remains small (0.004, col. 1) and becomes clearly negative once controls are introduced (about -0.008 across cols. 4–8). Although the adjusted R^2 rises more substantially in this window ($0.000 \rightarrow \sim 0.145$), the improvement is generated almost entirely by the control variables rather than by XRP itself. This suggests that any raw post-2020 comovement between XRP and Indian equities is effectively absorbed by standard macro financial shocks once they are included in the regression, leaving no robust residual loading that would signal an equity-like role for FX adjusted XRP in the Indian market.

Using the square-root variance ratio $\sqrt{\rho}$, both periods lie near unity ($\sqrt{\rho} \approx 1.019$ pre-2020; ≈ 1.023 post-2020), indicating relatively stable residual parity across samples and only a slight widening of the wedge later on. In combination, the full set of XRP results reinforces the broader conclusion for India that, even after 2020, FX adjusted crypto returns contribute little systematic information beyond what is already captured by global and domestic risk factors.

India's fully controlled post-2020 results (column 8) line up squarely with the emerging-market narrative of a weak, uneven crypto-FX channel rather than a tight equity-like loading. Once standard global risk factors and term-structure variables are included, the BTC coefficient is slightly negative (-0.007), ETH is small (0.006), LTC and XRP remain negative (-0.005 and -0.008), and USDT loads strongly negatively at -1.273 , pointing to a dominant currency-risk/flight-to-stablecoin mechanism rather than pro-equity exposure. The rise in adjusted R^2 into the 0.14 – 0.23 range is almost entirely delivered by the dollar, VIX, and rates, not by the crypto terms, so there is no coherent, positive, cross-asset pattern comparable to the disciplined equity-like comovement observed in developed markets. Instead, India's post-2020 configuration is fragmented and FX-driven, providing among the weakest support in the sample for a stable equity-like crypto-equity linkage.

4.11 South Africa

In this subsection, we study whether USD-denominated crypto assets, once converted into South African rand, move systematically with South African equity markets.

Tables 71 and 72 report the Bitcoin regressions for the pre-2020 and post-2020 windows, with the full set of progressive controls for commodities, the dollar, implied volatility, interest rates, and the South African term structure.

In the pre-2020 sample (Table 71), the bivariate slope on $\Delta \log(\text{BTC}_{ZAR})$ is effectively zero (about 0.000 in col. 1) and remains economically negligible once the full set of controls is included (about 0.002 in cols. 4–6, edging down to roughly 0.001 in cols. 7–8). Over the same sequence of specifications, the adjusted R^2 rises from almost none to approximately 0.201 . This pattern is consistent with a low comovement environment in which broad macro factors such as commodity prices, dollar movements, volatility, and interest rate changes explain the bulk of equity variation. In this setting, the parity wedge remains relatively large, and the Bitcoin leg does not load tightly on the pricing kernel relevant for South African equities. Any apparent association in the raw data is largely absorbed by global or dollar centered shocks.

Table 71: Pre-2020 South Africa Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{ZAR})$	-0.000 (0.002)	0.000 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.003)	0.020*** (0.003)	0.020*** (0.003)	0.020*** (0.003)
$\Delta 1\text{M Yield}$						0.004 (0.006)	0.004 (0.006)	0.004 (0.006)
$\Delta 10\text{Y Spread}$							-0.000 (0.002)	-0.000 (0.002)
$\Delta 1\text{Y Spread}$								-0.001 (0.001)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	-0.000	0.101	0.134	0.192	0.202	0.202	0.202	0.201

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 72: Post-2020 South Africa Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{ZAR})$	0.053*** (0.007)	0.042*** (0.007)	0.036*** (0.006)	0.017*** (0.006)	0.016*** (0.006)	0.016*** (0.006)	0.021*** (0.006)	0.021*** (0.006)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.019*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10\text{Y Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.025	0.133	0.272	0.321	0.330	0.330	0.371	0.371

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Post-2020 (Table 72), the estimated relationship strengthens considerably. The bivariate coefficient on $\Delta \log(\text{BTC}_{ZAR})$ is large and highly significant (0.053 in col. 1). Once controls are introduced, the coefficient decreases but remains economically meaningful, settling around 0.016 in cols. 4–6 and rising modestly to about 0.021 once South African term structure variables enter in cols. 7–8. Model fit also improves markedly, with the adjusted R^2 increasing from 0.025 in the bivariate specification to roughly 0.371 in the fully controlled regression. This shift is consistent with a post 2020 environment in which crypto returns, after exchange rate conversion, load more directly on the forces that drive South African equities. As crypto markets deepen and integrate more tightly with global dollar cycles, shocks to Bitcoin increasingly coincide with episodes of stronger global risk appetite and looser dollar conditions, both of which lift South African equity valuations.

Using the square root variance ratio $\sqrt{\rho}$, both windows lie close to unity ($\sqrt{\rho} \approx 1.006$ pre 2020; ≈ 1.015 post 2020). Although the post 2020 value is slightly larger, indicating a marginally wider residual gap, both figures imply that the unexplained wedge is small relative to idiosyncratic crypto variation. The key distinction across periods therefore lies not in residual dispersion but in the strength and persistence of the Bitcoin loading. Pre 2020, the crypto leg offered little incremental information once global factors were accounted for. After 2020, the positive and robust slope suggests a more stable channel through which crypto shocks transmit into South African equities, consistent with a shift from a largely hedge like behavior to modest but sustained positive co movement.

The results for Ethereum in Tables 73 and 74 follow the same broad pattern documented for Bitcoin, although the magnitudes differ. In the pre-2020 sample, the Ethereum coefficient is small in the bivariate specification (0.005, col. 1) and declines to an even more modest value of about 0.003 once standard controls are introduced (cols. 4–8). Over the same sequence of specifications, the adjusted R^2 rises from 0.001 to roughly 0.142, showing that broad macro financial forces such as commodities, the dollar, volatility, interest rates, and the term structure account for most of the variation in South African equity returns before 2020. In this environment, the FX adjusted ETH return carries little systematic information beyond what is already captured by global and domestic risk conditions. This is consistent with the model's interpretation of a wide parity wedge in the early period, where fragmented crypto markets and limited integration with global dollar funding prevent the Ethereum leg from loading meaningfully on the pricing kernel relevant for South African equity prices.

Table 73: Pre-2020 South Africa Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{ZAR})$	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.004*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10Y \text{ Yield}$					0.019*** (0.007)	0.019*** (0.007)	0.019*** (0.007)	0.019*** (0.007)
$\Delta 1M \text{ Yield}$						0.008 (0.008)	0.008 (0.008)	0.008 (0.008)
$\Delta 10Y \text{ Spread}$							0.007 (0.005)	0.007 (0.005)
$\Delta 1Y \text{ Spread}$								0.000 (0.000)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.001	0.075	0.091	0.136	0.141	0.141	0.142	0.142

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 74: Post-2020 South Africa Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{ZAR})$	0.047*** (0.005)	0.038*** (0.005)	0.030*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.016*** (0.005)	0.016*** (0.005)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10Y \text{ Yield}$					0.020*** (0.004)	0.019*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1M \text{ Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10Y \text{ Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1Y \text{ Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.035	0.139	0.275	0.321	0.331	0.331	0.370	0.370

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post-2020 sample, the Ethereum coefficient strengthens noticeably in the bivariate specification (0.047, col. 1) and, although reduced by the full set of controls, remains economically meaningful at approximately 0.014 in cols. 4–6 and 0.016 once term structure variables are included (cols. 7–8). The adjusted R^2 improves from 0.035 to roughly 0.370, indicating that a portion of Ethereum’s raw correlation with South African equities is absorbed by common global shocks but not entirely eliminated. This pattern mirrors the structural shift observed for Bitcoin: after 2020, the Ethereum leg becomes more aligned with the global dollar risk environment, and its FX-adjusted return increasingly comoves with domestic equities when global

risk appetite strengthens or weakens.

The square root variance ratio remains close to unity in both windows ($\sqrt{\rho} \approx 1.003$ before 2020 and ≈ 1.006 after 2020), indicating that the idiosyncratic wedge remains small relative to the idiosyncratic crypto component. The slight increase in $\sqrt{\rho}$ in the post-2020 period is consistent with a narrower parity gap and with the broader transition described in the model, in which deeper market integration and greater dollar cycle sensitivity cause FX-adjusted Ethereum returns to move more closely with the South African equity pricing kernel. Overall, the South African evidence points to a notable strengthening of Ethereum's equity-like behavior after 2020, even though much of the interaction continues to be shaped by global and domestic risk factors.

A broadly similar pattern of limited incremental explanatory power is evident for Litecoin. The pre-2020 regressions show weak and imprecise comovement: the Litecoin slope is effectively zero in the bivariate specification (-0.003 , col. 1) and remains economically negligible once common shock controls are introduced, stabilizing around ≈ 0.001 in cols. 4–8. Model fit rises from essentially nothing to a modest level (Adj. R^2 : $-0.001 \rightarrow 0.158$), which is consistent with the view that broad macroeconomic forces such as commodities, the dollar, volatility, interest rates, and the term structure account for most of the variation in South African equities before 2020. In this environment, it $\Delta \log(\text{LTC}_{ZAR})$ adds virtually no additional information, a pattern that is natural in a market where crypto activity remained shallow and only loosely connected to the domestic pricing kernel.

Table 75: Pre-2020 South Africa Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{ZAR})$	-0.003 (0.006)	-0.001 (0.006)	0.001 (0.006)	0.001 (0.005)	0.001 (0.005)	0.001 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.003*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.015 (0.009)	0.014 (0.009)	0.013 (0.009)	0.013 (0.009)
$\Delta 1\text{M Yield}$						0.012 (0.012)	0.011 (0.012)	0.011 (0.012)
$\Delta 10\text{Y Spread}$							0.009 (0.007)	0.009 (0.007)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.080	0.120	0.156	0.158	0.158	0.158	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In the post 2020 period, the Litecoin coefficient becomes clearly positive and significant in the bivariate regression (0.029, col. 1), indicating that raw comovement with South African equities increases once USD denominated Litecoin returns are converted into rand. However, this loading attenuates sharply when common shocks are absorbed, falling to about 0.007 in cols. 4–6 and rising modestly to ≈ 0.010 when term structure variables are included in cols. 7–8. At the same time, the adjusted R^2 improves substantially, moving from 0.017 to roughly 0.368. These results suggest that much of the observed post 2020 comovement is driven by the same global and domestic macroeconomic factors that shape the broader risk environment, leaving Litecoin with a modest but positive residual loading after controls.

Table 76: Post-2020 South Africa Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{ZAR})$	0.029*** (0.005)	0.023*** (0.005)	0.020*** (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)	0.010** (0.004)	0.010** (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.020*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10\text{Y Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.017	0.127	0.268	0.319	0.329	0.329	0.368	0.368

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Using the square root variance ratio $\sqrt{\rho}$, both windows lie close to one ($\sqrt{\rho} \approx 1.009$ pre 2020; ≈ 1.007 post 2020), indicating only a slightly looser parity in the early sample and reinforcing the interpretation that the dominant shift after 2020 reflects stronger common shock driven co movement rather than a large and independent Litecoin equity linkage. Together, these findings support the emerging market narrative in which crypto assets exhibit uneven and relatively weak transmission once macroeconomic conditions are properly accounted for.

The pre-2020 regressions in Table 77 show a modest and mostly negative bivariate loading of Tether on South African equities (-0.102 , col. 1) that is rapidly absorbed once common shocks are controlled for. As commodities, the dollar, volatility, interest rates, and term structure variables are added, the USDT coefficient shrinks toward zero and becomes statistically weak (cols. 3–8), while the adjusted R^2 rises from 0.025 to approximately 0.158. This pattern is consistent with a pre 2020 hedge or low comovement regime, in which broad macro forces explain most equity variation and $\Delta \log(\text{USDT}_{ZAR})$ carries little incremental information beyond these standard risk factors.

Table 77: Pre-2020 South Africa Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{ZAR})$	-0.102*** (0.021)	-0.065*** (0.021)	-0.032 (0.022)	-0.015 (0.021)	-0.015 (0.021)	-0.015 (0.021)	-0.020 (0.022)	-0.020 (0.022)
$\Delta \text{Commodity}$		0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.014* (0.008)	0.014* (0.008)	0.013* (0.008)	0.013* (0.008)
$\Delta 1\text{M Yield}$						0.005 (0.010)	0.005 (0.010)	0.005 (0.010)
$\Delta 10\text{Y Spread}$							0.009 (0.006)	0.009 (0.006)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.025	0.091	0.113	0.156	0.158	0.157	0.158	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 78: Post-2020 South Africa Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{ZAR})$	-0.421*** (0.028)	-0.322*** (0.028)	-0.084*** (0.030)	0.023 (0.030)	0.011 (0.030)	0.010 (0.030)	0.107*** (0.030)	0.107*** (0.030)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.011*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.019*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.002 (0.003)	0.002 (0.003)
$\Delta 10\text{Y Spread}$							-0.029*** (0.002)	-0.029*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.099	0.170	0.263	0.319	0.328	0.328	0.370	0.370

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

In the post-2020 sample, summarized in Table 78, the raw bivariate effect is large and negative (-0.421 , col. 1), but this apparent disconnection between Tether and South African equities is again strongly mediated by macro controls. Once the same set of common shock proxies is included, the USDT slope attenuates and eventually flips sign, settling at a small but positive and statistically significant loading (around 0.107 in cols. 7–8), while the adjusted R^2 improves from 0.099 to roughly 0.370 . From an economic perspective, this evolution is in line with a stablecoin that primarily transmits dollar and funding conditions: the raw negative response reflects periods when flight to USDT coincides with local equity stress, whereas the fully controlled specification isolates a residual, mildly pro equity exposure that emerges only after the main macro channels have been netted out. Using the square-root variance ratio $\sqrt{\rho}$, parity loosens from $\sqrt{\rho} \approx 1.204$ in the pre

2020 window to ≈ 1.630 after 2020, indicating a larger residual gap in the later period and reinforcing the interpretation that any post 2020 USDT equity linkage in ZAR terms is small, fragile, and heavily conditioned by the broader macro environment.

Finally, the pre-2020 regressions in Table 79 indicate essentially no systematic relation between Ripple returns and South African equities. The bivariate XRP slope is indistinguishable from zero in column (1) (0.000), and once controls for commodities, the dollar, implied volatility, interest rates, and the term structure are introduced, the coefficient remains economically small, drifting to approximately 0.005 by columns (4) to (8). Over this progression, the adjusted R^2 rises from -0.001 to about 0.159, indicating that the explanatory gains come almost entirely from standard macrofinancial forces rather than from $\Delta \log(\text{XRP}_{ZAR})$. This pattern is consistent with a low comovement environment in which broad global and domestic risk factors dominate the pricing of South African equities, leaving the FX-converted XRP leg with negligible incremental power.

Table 79: Pre-2020 South Africa Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{ZAR})$	0.000 (0.006)	0.003 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.003*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10Y \text{ Yield}$					0.015 (0.009)	0.014 (0.009)	0.013 (0.009)	0.013 (0.009)
$\Delta 1M \text{ Yield}$						0.012 (0.012)	0.011 (0.012)	0.011 (0.012)
$\Delta 10Y \text{ Spread}$							0.008 (0.007)	0.008 (0.007)
$\Delta 1Y \text{ Spread}$								0.000 (0.000)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.081	0.121	0.157	0.158	0.158	0.159	0.159

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post2020 regressions in Table 80 show somewhat stronger unconditional association, but most of that linkage is absorbed by common shocks once controls enter. The bivariate coefficient is clearly positive and significant (0.016 in column (1)), suggesting that Ripple returns moved more in tandem with South African equities during this period. However, once the full set of macro controls is added, the slope attenuates and stabilizes around a small positive loading of roughly 0.005 in columns (4) to (6), rising modestly to about 0.007 when term spreads are included in columns (7) and (8). Model fit improves accordingly, with the adjusted R^2 increasing from 0.006 in the bivariate specification to approximately 0.368 in the fully controlled version. This progression suggests that while raw comovement between Ripple and South African equities strengthened after 2020, much of this relation reflects exposure to shared global and domestic shocks rather than an independent contribution from the crypto asset itself.

Using the square root variance ratio, $\sqrt{\rho}$, both subsamples show values extremely close to unity, with $\sqrt{\rho} \approx 1.007$ before 2020 and ≈ 1.005 after 2020. These figures indicate that the residual parity gap remains very small relative to the idiosyncratic crypto component in both windows, and that any tightening in parity post2020 is modest. Taken together, the results imply that XRP's apparent post2020 equity comovement is largely mediated by common macro forces, leaving only a limited and residual positive loading on South African equities once these forces are netted out.

Table 80: Post-2020 South Africa Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{ZAR})$	0.016*** (0.004)	0.012*** (0.004)	0.012*** (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.007** (0.003)	0.007** (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10Y \text{ Yield}$					0.020*** (0.004)	0.020*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1M \text{ Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10Y \text{ Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1Y \text{ Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.006	0.120	0.264	0.319	0.328	0.329	0.368	0.368

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

South Africa, focusing on the fully controlled specifications (column 8), shows a clean transition from negligible pre-2020 linkages to a pronounced, broad-based post-2020 equity-like crypto loading. The BTC coefficient rises from 0.001 to 0.021, ETH from 0.003 to 0.016, LTC from 0.001 to 0.010, XRP from 0.005 to 0.007, and USDT flips from -0.020 to 0.107 . Over the same shift, the column 8 adjusted R^2 values increase from roughly $0.16 - 0.20$ to a tight $0.36 - 0.37$ band across all five assets, indicating that once standard global factors (commodities, dollar, volatility, term structure) are controlled for, South African equities load consistently and positively on FX-adjusted crypto returns. Among the emerging markets in our sample, this transition makes South Africa the clearest case where the post-2020 crypto–FX channel resembles the stable, equity-like comovement seen in advanced economies, while simultaneously highlighting that, in the cross-section of EMs, the channel remains heterogeneous and fragmented rather than governed by a single universal loading. In the next section, we shift from static cross-sectional regressions to a dynamic perspective, examining how the strength of the crypto–equity linkage evolves over time through rolling-window correlations and formal hypothesis testing on short-run betas across countries.

5 Conclusion

This paper has examined how USD-denominated crypto assets transmit into domestic equity markets through the exchange rate, and how this crypto–FX–equity linkage differs across developed and emerging economies. We began from a simple but empirically motivated observation: Bitcoin has transitioned from an asset with near-zero or even negative comovement with equities to one that behaves like a high-beta equity factor. Rolling one-year correlations and volatility wedges show three distinct phases—an early period of weak or negative comovement with large wedges, a transition phase with compression of volatility wedges and rising correlations, and a post-2019 regime in which correlations remain persistently positive and elevated. These reduced-form facts suggest that crypto has moved from behaving as an orthogonal “digital gold” to loading on similar risk as local equities.

To rationalize these patterns, we built a tractable portfolio-choice model in which a representative domestic investor allocates between a local equity claim and a USD-priced crypto asset, repatriating all wealth into local currency each period. Expressing payoffs in real local terms and imposing standard Euler equations, we derived a risk-adjusted parity between domestic equities and the FX-converted crypto leg. This parity gives rise to a *wedge*—the difference between real equity returns and real FX-adjusted crypto returns—whose conditional mean is entirely driven by the interaction between the stochastic discount factor and the return

differential. Proposition 1 formalizes this wedge decomposition, while Proposition 2 links the short-run regression slope of domestic equities on the FX-adjusted crypto leg to the variance of an idiosyncratic parity wedge. Once common shocks are controlled for, the model implies a strictly positive slope that concentrates near one whenever the wedge variance is small relative to the volatility of the crypto leg; equivalently, the slope is tightly tied to the partial correlation between the two legs.

The empirical analysis supports these implications and reveals a sharp post-2020 regime shift. Using daily data for the United States, we show that pre-2020, once standard global controls (commodities, dollar index, volatility, and interest rates) are included, Bitcoin, Ethereum, Litecoin, and Ripple contribute little incremental explanatory power for U.S. equity returns: their coefficients are small, often unstable in sign, and statistically weak, while the regression fit rises mainly because common shocks are absorbed. After 2020, the picture reverses. The slopes on the major volatile tokens are large and highly significant in bivariate regressions and remain economically meaningful once controls are added, with adjusted R^2 values increasing substantially. By contrast, Tether—a USD-pegged stablecoin—displays coefficients that are indistinguishable from zero in both subsamples, and its wedge statistics are dominated by peg mechanics rather than genuine equity risk. Together, these results show that non-pegged USD crypto assets now load on U.S. equities in an equity-like fashion, whereas a pure dollar proxy does not, exactly as predicted by the crypto-FX-equity framework. This pattern is echoed, with local nuances, in the other large developed markets in our sample.

Extending the analysis across countries, we find a systematic developed–emerging market divergence. In major developed markets, FX-adjusted crypto returns and local equity indices comove strongly and persistently in the post-2020 period, and the estimated slopes β_i remain positive and often sizeable even after controlling for macro and financial factors. This is consistent with a tight parity wedge and a dominant “crypto channel” in which crypto behaves as an additional equity factor. By contrast, in emerging markets, the regressions offer weaker and less robust support for a direct crypto–equity loading: conditional slopes are smaller on average and frequently lose significance once FX and global variables are included, and the wedge statistics point to a larger, more volatile parity gap. South Africa is a notable exception. Its post-2020 fully controlled estimates display broad-based, positive and economically non-trivial betas on all five assets—with adjusted R^2 values rising into the 0.36–0.37 range—closely resembling the developed-market pattern. The evidence therefore suggests that, outside a handful of more financially integrated cases such as South Africa, transmission in emerging economies operates primarily through the exchange-rate leg and country-specific frictions—capital controls, liquidity constraints, and policy interventions—rather than through a single, tight loading on crypto itself. The state-dependent wedge framework provides a unified way to interpret these asymmetries: where market integration is high and frictions limited, wedge variance is compressed and the slope is pushed toward an equity-like loading; where frictions are sizable, wedge variance remains large and comovement is weaker or episodic.

Our findings have direct implications for both portfolio management and policy. For global investors, the post-2020 evidence implies that treating Bitcoin and other major tokens as diversifiers or “digital gold” is increasingly misleading in developed markets: they now behave as high-beta extensions of equity exposure once FX conversion is taken into account. For policymakers in emerging markets, the results highlight that crypto is not only a speculative side market but an additional channel through which dollar risk and global financial conditions can permeate domestic asset prices—either directly through crypto positions or indirectly via the exchange rate and on/off-ramp frictions. More broadly, the paper shows that a simple SDF-based parity, combined with a measurable parity wedge, can reconcile the evolution of raw correlations with a structural account of how crypto and equities are jointly priced. Future work could extend the framework to richer forms of heterogeneity, multiple crypto assets and stablecoins, or settings with explicit capital controls and liquidity freezes, but the core message is robust: as long as both legs load on the same pricing kernel, compressing wedge variance turns crypto from an apparent hedge into a conduit that transmits, rather than offsets, equity risk.

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Appendix A. Proofs and Further Theoretical Results

This appendix provides proofs for the model implications in Section 3 and the formal justification for Propositions 1 and 2. Throughout, expectations are conditional on $\mathcal{F}_t$ and all returns are *real* unless otherwise noted. We assume interior solutions ($n_t > 0$, $A_t^k > 0$) and that $\{C_t\}$ is strictly positive and adapted.

A.1. Stochastic discount factor and parity

Lemma 1 (SDF in real LCU and Euler equations). *Define the nominal LCU stochastic discount factor*

$$m_{t+1}^N := \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{P_t^c}{P_{t+1}^c} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{1}{1 + \pi_{t+1}}.$$

Then any tradable payoff $1 + R_{t+1}$ quoted in nominal LCU satisfies

$$\mathbb{E}_t[m_{t+1}^N(1 + R_{t+1})] = 1.$$

Equivalently, for the associated real gross return

$$1 + R_{t+1}^{\text{real}} := \frac{1 + R_{t+1}}{1 + \pi_{t+1}} = \frac{P_t^c}{P_{t+1}^c}(1 + R_{t+1}),$$

the real LCU SDF

$$m_{t+1}^R := \beta \frac{u'(C_{t+1})}{u'(C_t)}$$

satisfies

$$\mathbb{E}_t[m_{t+1}^R(1 + R_{t+1}^{\text{real}})] = 1.$$

Proof. From the Bellman problem in §3.3, the Lagrange multiplier on the period- t budget constraint is $\lambda_t = u'(C_t)/P_t^c$. FOC for n_t and A_t^k give, respectively,

$$\lambda_t P_t^s = \beta E_t[\lambda_{t+1}(P_{t+1}^s + D_{t+1})], \quad \lambda_t = \beta E_t \left[\lambda_{t+1} \frac{S_{t+1}}{S_t} (1 + \Theta_{t+1}^*) \right].$$

Divide by λ_t , substitute $\lambda_{t+1}/\lambda_t = (u'(C_{t+1})/u'(C_t))(P_t^c/P_{t+1}^c)$, and rearrange to obtain

$$\mathbb{E}_t[m_{t+1}^N(1 + R_{t+1}^s)] = 1, \quad \mathbb{E}_t[m_{t+1}^N(1 + R_{t+1}^k)] = 1.$$

Using $1 + R_{t+1}^{\text{real}} = (P_t^c/P_{t+1}^c)(1 + R_{t+1})$ and $m_{t+1}^R = m_{t+1}^N (P_{t+1}^c/P_t^c)$ gives the real versions. $\square$

Lemma 2 (Risk-adjusted parity). *Let $m_{t+1} := m_{t+1}^R = \beta u'(C_{t+1})/u'(C_t)$ denote the real LCU SDF. For the domestic equity and FX-converted crypto legs in real LCU,*

$$E_t[m_{t+1}((1 + R_{t+1}^{s,\text{real}}) - (1 + R_{t+1}^{k,\text{real}}))] = 0.$$

Equivalently, letting $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$,

$$E_t[w_{t+1}] = -\frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} . \tag{6}$$

A.2. Proof of Proposition 1 (wedge decomposition)

Proof. By Lemma 2, for $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$, and use the identity that

$$\mathbb{E}_t[m_{t+1}w_{t+1}] = \mathbb{E}_t[m_{t+1}]\mathbb{E}_t[w_{t+1}] + \text{Cov}_t(m_{t+1}, w_{t+1}),$$

the parity condition implied that we have

$$E_t[w_{t+1}] = -\frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]}.$$

Decompose the wedge w_{t+1} around its conditional mean:

$$w_{t+1} = E_t[w_{t+1}] + \varepsilon_{t+1}, \quad E_t[\varepsilon_{t+1}] = 0. \quad (7)$$

$$\varepsilon_{t+1} := w_{t+1} - E_t[w_{t+1}]. \quad (8)$$

Since

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} + w_{t+1},$$

substituting the decomposition of w_{t+1} yields

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \quad E_t[\varepsilon_{t+1}] = 0.$$

This is exactly the representation stated in Proposition 1: the nonzero conditional mean of the wedge is fully driven by the SDF–wedge interaction $\text{Cov}_t(m_{t+1}, w_{t+1})$. In the special case $\text{Cov}_t(m_{t+1}, w_{t+1}) = 0$, the conditional means of the two real returns coincide. $\square$

A.3. Proof of Proposition 2 (short-run slope $> 0, \approx 1$)

We restate the empirical projection (with time fixed effects as controls for common shocks):

$$r_{t+1}^{s,\text{real}} = \alpha + \beta r_{t+1}^{k,\text{real}} + \lambda_{t+1} + u_{t+1}, \quad E[r_{t+1}^{k,\text{real}} \lambda_{t+1}] = 0, \quad E[u_{t+1}(\cdot)] = 0.$$

Proof. Let a *single-factor* decomposition capture the shocks common to the two legs:

$$\begin{aligned} r_{t+1}^{s,\text{real}} &= \mu_s + b' F_{t+1} + \varepsilon_{t+1}^s, \\ r_{t+1}^{k,\text{real}} &= \mu_k + b'_k F_{t+1} + \varepsilon_{t+1}^k, \end{aligned}$$

where F_{t+1} stacks common risk shocks (including FX and global conditions) and the idiosyncratic components $\varepsilon_{t+1}^s, \varepsilon_{t+1}^k$ are mean zero and orthogonal to F_{t+1} . Time fixed effects λ_{t+1} span the space of F_{t+1} (up to estimation noise), so the Frisch–Waugh–Lovell theorem implies that the coefficient β in the regression of $r_{t+1}^{s,\text{real}}$ on $r_{t+1}^{k,\text{real}}$ *controlling for* λ_t equals the slope in the auxiliary regression of the residuals:

$$\underbrace{(r_{t+1}^{s,\text{real}} - \mu_s - b' F_{t+1})}_{\varepsilon_{t+1}^s} = \beta \underbrace{(r_{t+1}^{k,\text{real}} - \mu_k - b'_k F_{t+1})}_{\varepsilon_{t+1}^k} + u_{t+1}.$$

Hence,

$$\beta = \frac{\text{Cov}(\varepsilon_{t+1}^s, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)} = 1 - \frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{2\text{Var}(\varepsilon_{t+1}^k)} + \frac{\text{Var}(\varepsilon_{t+1}^s) - \text{Var}(\varepsilon_{t+1}^k)}{2\text{Var}(\varepsilon_{t+1}^k)}. \quad (9)$$

$$\beta = \frac{\text{Cov}(\varepsilon_{t+1}^s, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)} = 1 + \frac{\text{Cov}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}. \quad (10)$$

By Cauchy–Schwarz,

$$|\text{Cov}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k, \varepsilon_{t+1}^k)| \leq \sqrt{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k) \text{Var}(\varepsilon_{t+1}^k)}.$$

Equivalently, the two-sided bound is

$$-\sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}} \leq \frac{\text{Cov}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)} \leq \sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}.$$

Thus,

$$1 - \sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}} \leq 1 + \frac{\text{Cov}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)} \leq 1 + \sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}.$$

Substituting into

$$\beta = 1 + \frac{\text{Cov}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k, \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)},$$

we obtain the bound

$$1 - \sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}} \leq \beta \leq 1 + \sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}.$$

In the absence of large state-dependent risk adjustments, we have that $\varepsilon_{t+1}^s \approx \varepsilon_{t+1}^k$ and $\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)$ is small. Therefore, substitute to obtain

$$\beta = 1 + O\left(\sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}\right),$$

so $\beta \approx 1$ and, in particular, $\beta > 0$ whenever $\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k) < \text{Var}(\varepsilon_{t+1}^k)$. Alternatively,

$$\beta = 1 - O\left(\sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}\right),$$

so $\beta \approx 1$ and, in particular, $\beta > 0$ whenever $\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k) < \text{Var}(\varepsilon_{t+1}^k)$. Note

$$O\left(\sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}\right) \rightarrow 0 \quad \text{as} \quad \left(\sqrt{\frac{\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k)}{\text{Var}(\varepsilon_{t+1}^k)}}\right) \rightarrow 0,$$

because $\text{Var}(\varepsilon_{t+1}^s - \varepsilon_{t+1}^k) \rightarrow 0$. Therefore $\beta > 0$ and $\beta \approx 1$ whenever the idiosyncratic discrepancy between the two legs is second-order relative to the variation in the crypto leg after removing common shocks. $\square$

Appendix B. Descriptive Evidence on Bitcoin–Equity Co-movement

This appendix reports additional descriptive evidence and provides motivation on the joint behavior of Bitcoin and local equity markets. The figures present **unconditional** patterns; the developed–emerging market asymmetries discussed in the main text are based on **conditional** regression evidence that controls for exchange-rate and global risk factors.

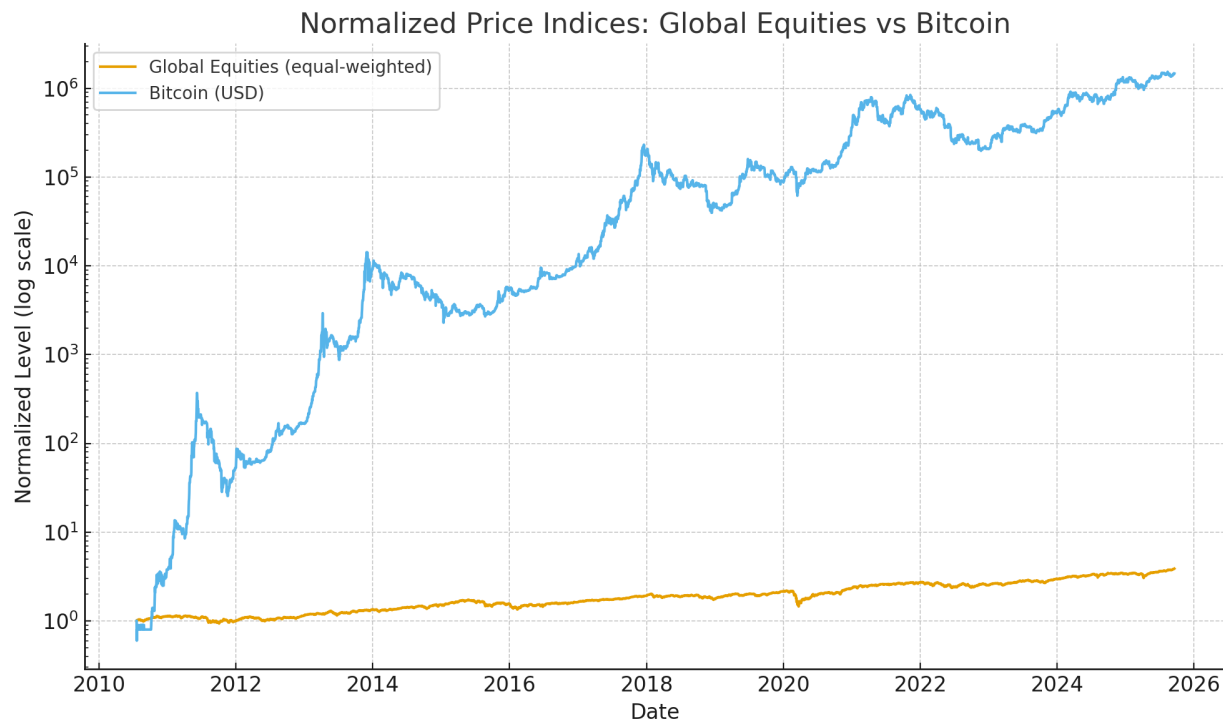


Figure 2: **Normalized price indices: global equities vs. Bitcoin (unconditional).** The figure plots normalized price indices for an equal-weighted average of the eight local equity indices in our sample and for Bitcoin in USD from 2011 to 2025. Both series are normalized to one at the start of the sample; the vertical axis for Bitcoin is on a logarithmic scale. The chart illustrates two facts that motivate the wedge-based model: (i) Bitcoin’s level and volatility are orders of magnitude larger than those of global equities, implying a substantial idiosyncratic crypto component that any parity relation must absorb through a wedge; and (ii) especially after 2020, the major booms and busts in Bitcoin and global equities become more synchronized, foreshadowing the tighter, equity-like comovement documented in our conditional regressions and rolling-window estimates.

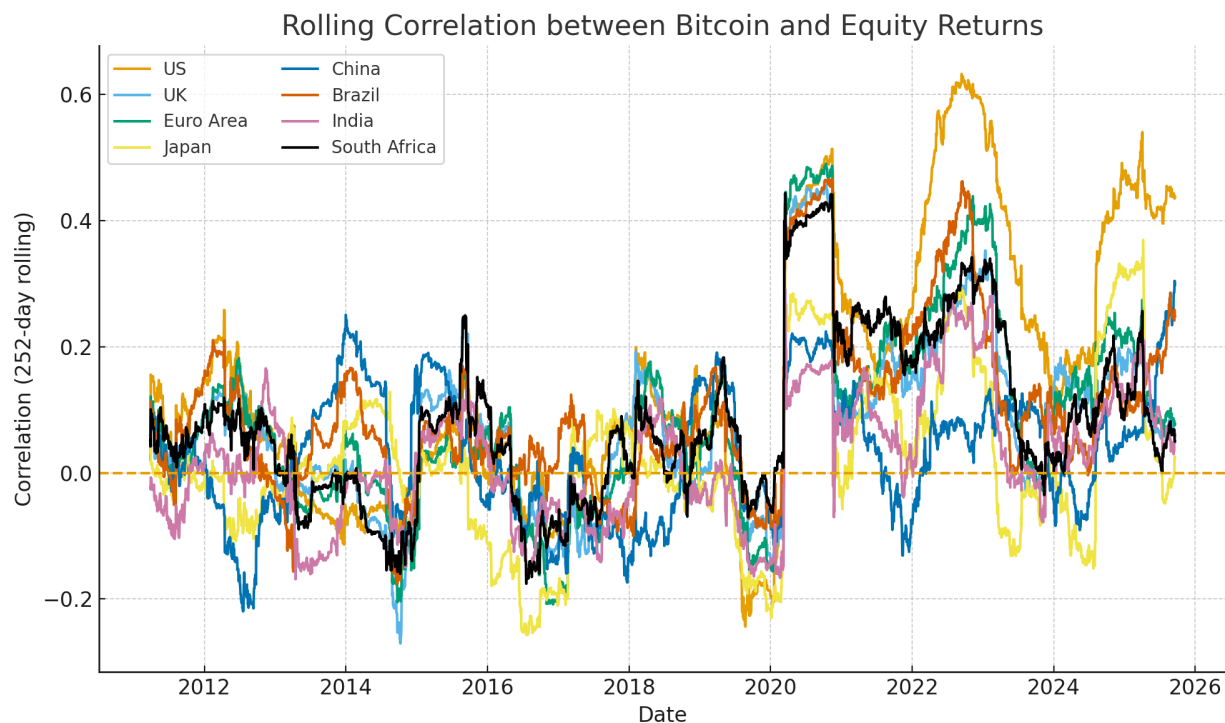


Figure 3: **Rolling correlation between Bitcoin and equity returns (unconditional).** For each country, the figure shows the one-year (252 trading days) rolling correlation between daily log returns on Bitcoin (USD) and the local equity index. Correlations were close to zero or slightly negative in the early part of the sample and become persistently positive after 2020 for most countries. The series exhibit time-varying comovement, with episodes of elevated correlation around major global risk events and crypto booms, consistent with the regression evidence that equity–crypto linkages have strengthened over time, especially in developed markets. These patterns are purely descriptive and do not condition on exchange-rate movements or macro variables.

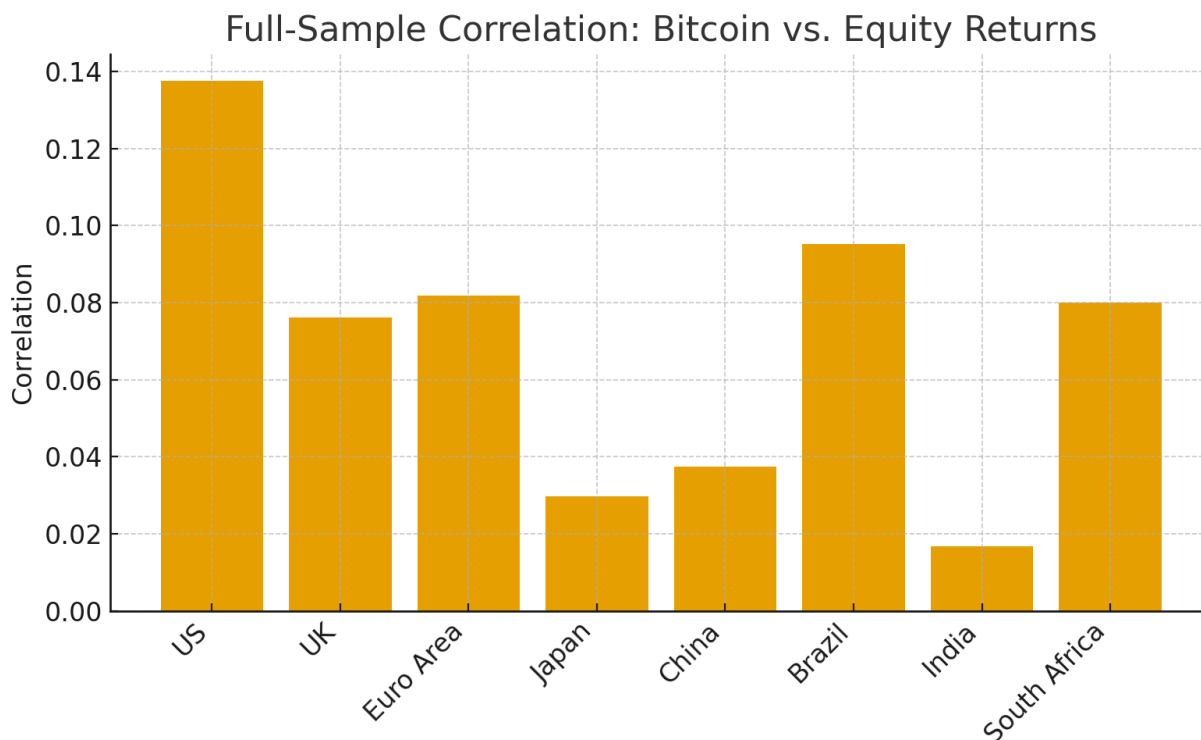


Figure 4: **Full-sample unconditional correlation: Bitcoin vs. equity returns.** The bar chart reports full sample, unconditional correlations between daily Bitcoin returns (in USD) and local equity returns for the eight countries. Correlations are modestly positive and display cross-country heterogeneity. Cross-country differences in these unconditional correlations are informative but do not account for exchange-rate movements or global controls; the developed–emerging market divergence discussed in the main text is established using conditional regressions that explicitly separate the direct “crypto channel” from the FX transmission channel. Even so, these unconditional correlations support the narrative: developed markets exhibit a noticeably higher aggregate correlation (≈ 0.33) than emerging markets (≈ 0.23), consistent with tighter integration and a stronger underlying crypto–equity channel in advanced economies.